Two-Timescale Secure Transmission for Movable Antenna-Enabled Multiuser Systems With Statistical Eavesdropper CSI

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***Abstract*—Movable antennas (MAs) can reshape the spatial correlation between the channels of legitimate users (LUs) and an eavesdropper (Eve), thereby providing a new degree of freedom for physical layer security. However, existing MA-aided secure designs rely on instantaneous channel state information (CSI), which is hindered by the antenna movement delay, the CSI acquisition overhead over the movable region, and the unavailability of the instantaneous CSI of a passive Eve. To overcome these limitations, we propose a two-timescale secure transmission framework for an MA-enabled multiuser downlink system. Specifically, the antenna positions and the power allocation between the confidential signals and artificial noise (AN) are optimized based on statistical CSI, while zero-forcing precoding and null-space AN are designed based on the instantaneous CSI of the LUs only. We prove that the information leakage and the AN received by Eve are conditionally independent given the channels of the LUs, and derive a closed-form approximation of the ergodic wiretap rate, which reveals how the antenna positions reduce the information leakage and increase the AN received by Eve. Then, we develop an alternating optimization algorithm to maximize the approximate ergodic secrecy sum rate, where the optimal power allocation follows a secrecy water-filling structure, and each antenna position is updated by a global grid search followed by majorization–minimization refinement. Simulation results demonstrate the significant advantages of the proposed design over various benchmarks in terms of both the secrecy rate and the required AN power.**

***Index Terms*—Movable antenna, physical layer security, artificial noise, two-timescale design, statistical CSI, ergodic secrecy rate.**

I. Introduction

PHYSICAL layer security (PLS) exploits the randomness and spatial characteristics of wireless channels to protect confidential messages against eavesdropping, thereby complementing upper-layer cryptography [1]–[3]. In multi-antenna systems, PLS is mainly enabled by secure precoding and artificial noise (AN) [4], [5]. The former steers the confidential signals toward the legitimate users (LUs), whereas the latter degrades the channel of the eavesdropper (Eve) without interfering with the LUs.

The effectiveness of both techniques relies on the spatial separability between the LUs and Eve. However, when Eve is located in a direction close to that of an LU and the line-of-sight (LoS) propagation is dominant, their channels are highly correlated. In this case, a conventional fixed-position antenna (FPA) array can hardly steer the information-bearing beams away from Eve without sacrificing the beamforming gain at the LU. Movable antennas (MAs), also known as fluid antennas, have recently emerged as a promising technology to overcome this limitation [6]–[8]. By adjusting the antenna positions within a given region, MAs reshape the channel correlation among different receivers without additional radio-frequency (RF) chains [9]–[11]. This capability can be exploited to decorrelate the channels of the LUs from that of Eve. For example, the authors of [12] jointly optimized the transmit beamforming and the positions of an MA array to maximize the secrecy rate with perfect channel state information (CSI), and the authors of [13] exploited the statistical CSI of the eavesdroppers when their instantaneous CSI is unavailable. More recently, a deterministic equivalent of the ergodic secrecy rate was derived in [14] for a multiple-input multiple-output multiple-eavesdropper-antenna MA system with partial CSI of Eve, and a sensing-assisted secure design was developed in [15], where the MA positions enhance both the estimation of the direction of Eve and the secrecy rate.

Despite these advances, the aforementioned works optimize the antenna positions based on the instantaneous CSI of at least the LUs, which requires the antennas to be repositioned in each channel coherence interval (CI) under fast fading. Such single-timescale designs face three practical difficulties. First, the mechanical movement incurs non-negligible delay and energy consumption. Second, the CSI is required over the entire movable region, which, under Rician fading, can only be acquired by moving the antennas across the region [16], [17]. Third, the instantaneous CSI of a passive Eve is generally unavailable. To reduce the overhead, the two-timescale design, which was first developed for intelligent reflecting surfaces [18], has been extended to MA systems [16], [17]. In these two-timescale MA designs, the antenna positions are optimized based on the statistical CSI, and the beamforming is designed based on the instantaneous CSI of the resulting effective channels. However, these works focused on the ergodic sum rate, and the two-timescale design of MA-enabled secure transmission remains unexplored.

Extending the two-timescale MA design to secure transmission is non-trivial for three reasons. First, the ergodic wiretap rate of Eve is determined by the random precoding directions and AN subspace induced by the instantaneous CSI of the LUs, whereas the existing ergodic-rate approximations for zero-forcing (ZF) precoding [16], [17], [19] only characterize the signal power received by the intended users. Second, in the rate-oriented design, the antenna positions improve the ergodic rate only by decorrelating the users, and thus bring no gain

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for a single user [16], whereas in secure transmission they also determine the information leakage and the AN received by Eve. Third, the power allocation between the confidential signals and the AN is coupled with the antenna positions. To address these challenges, this paper makes the following main contributions.

- We propose a two-timescale secure transmission framework for MA-enabled multiuser systems. The antenna positions and the long-term power allocation between the confidential signals and AN are optimized based on the statistical CSI of the LUs and Eve, whereas ZF precoding and null-space AN are designed based on the instantaneous CSI of the LUs only. Hence, the framework requires neither the instantaneous CSI of Eve nor frequent antenna movement.
- We prove that the information leakage and the AN received by Eve are conditionally independent given the channels of the LUs. Based on this property, we derive a closed-form approximation of the ergodic wiretap rate, which captures the LoS correlation between Eve and the LUs and is accurate over a wide range of Rician factors. It reveals that, in contrast to the rate-oriented design, the MAs enhance secrecy by reducing the information leakage and increasing the AN received by Eve, even for a single LU.
- We develop an alternating optimization (AO) algorithm to maximize the resulting approximate ergodic secrecy sum rate (ESSR). For given antenna positions, the optimal power allocation is derived in a secrecy water-filling form. For a given power allocation, each antenna position is updated by a global grid search, which exploits a rank-one structure of the objective function, followed by majorization–minimization (MM) refinement. The algorithm converges monotonically and fast, and its complexity does not depend on the number of channel realizations.
- Simulation results demonstrate that the proposed design significantly outperforms the rate-oriented two-timescale MA design [16], antenna selection (AS) from twice as many antennas, and FPA arrays, while requiring considerably less AN power. The secrecy gain persists for a single LU, against an Eve that treats the inter-user interference as noise, and under moderate errors in the direction of Eve.

The rest of this paper is organized as follows. Section II presents the system model and problem formulation. Section III derives the approximate ESSR. Section IV develops the optimization algorithm. Section V provides simulation results, and Section VI concludes the paper.

Notations: Scalars, vectors, and matrices are denoted by italic, bold lower-case, and bold upper-case letters, respectively. $(\cdot)^T$, $(\cdot)^*$, and $(\cdot)^H$ denote the transpose, conjugate, and conjugate transpose, and $|\cdot|$, $\|\cdot\|$, and $\mathrm{tr}(\cdot)$ the modulus, Euclidean norm, and trace, respectively. $[\mathbf{x}]_i$ and $[\mathbf{X}]_{i,j}$ denote the i-th entry of $\mathbf{x}$ and the (i,j)-th entry of $\mathbf{X}$, $\mathrm{diag}(x_1,\ldots,x_M)$ a diagonal matrix, and $\mathbf{X}^{1/2}$ the entrywise square root of a nonnegative diagonal matrix $\mathbf{X}$. $\mathbf{I}_N$ denotes the $N\times N$ identity matrix, $\mathbf{e}_m$ the m-th column of an identity matrix, and $\mathcal{CN}(\boldsymbol{\mu},\boldsymbol{\Sigma})$ the circularly symmetric complex Gaussian distribution. $\mathbf{X}\succeq\mathbf{0}$ ($\mathbf{X}\succ\mathbf{0}$) means that $\mathbf{X}$ is positive semidefinite (definite). $j\triangleq\sqrt{-1}$, $\mathbb{E}\{\cdot\}$ and $\mathrm{Var}\{\cdot\}$ denote the expectation and variance, $\mathrm{Re}\{\cdot\}$ the real part, $[x]^+\triangleq\max\{x,0\}$, and $|\mathcal{S}|$ the cardinality of a set $\mathcal{S}$.

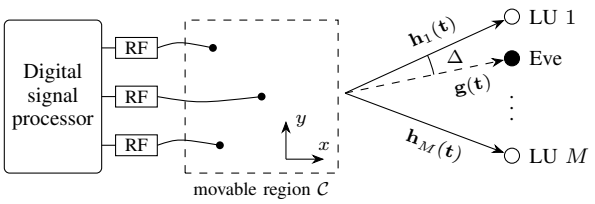


Fig. 1. An MA-enabled multiuser downlink system in the presence of a passive Eve, which is located in a direction close to that of LU 1 (angular offset Δ).

II. System Model and Problem Formulation

A. MA-Enabled Secure Multiuser Downlink System

As illustrated in Fig. 1, we consider the downlink of a multiuser system, where a base station (BS) equipped with N MAs serves M single-antenna LUs in the presence of a single-antenna passive Eve. Let $\mathcal{M}\triangleq\{1,\ldots,M\}$ and $\mathcal{N}\triangleq\{1,\ldots,N\}$ denote the index sets of the LUs and the MAs, respectively, where $N\geq M+1$ is required to transmit AN in the null space of the channels of the LUs. The MAs are connected to the RF chains via flexible cables and can move within a two-dimensional (2D) square region $\mathcal{C}\triangleq[-A/2,A/2]\times[-A/2,A/2]$ of side length A [6], [7]. In the local Cartesian coordinate system of $\mathcal{C}$, the position of the n-th MA is denoted by $\mathbf{t}_n\triangleq[x_n,y_n]^T\in\mathcal{C}$, and the positions of all the MAs are collected in $\mathbf{t}\triangleq[\mathbf{t}_1^T,\ldots,\mathbf{t}_N^T]^T\in\mathbb{R}^{2N\times 1}$. To avoid the coupling between adjacent antennas, the distance between any two MAs must be no smaller than a minimum distance $D_{\min}$.

Since the size of $\mathcal{C}$ is much smaller than the propagation distance, moving the MAs changes only the phases of the LoS paths, whereas their angles of departure (AoDs) and amplitudes remain unchanged [7]. Let $\theta_m\in[-\pi/2,\pi/2]$ and $\phi_m\in[-\pi/2,\pi/2]$ denote the elevation and azimuth AoDs of the LoS path from the BS to the m-th LU, respectively, and define $\mathbf{a}_m\triangleq[\cos\theta_m\sin\phi_m,\ \sin\theta_m]^T$. Then, the propagation distance of the LoS path from the n-th MA differs from that from the origin of $\mathcal{C}$ by $\mathbf{t}_n^T\mathbf{a}_m$, and the LoS component of the channel from the BS to the m-th LU is given by [16]

$$\bar{\mathbf{h}}_m(\mathbf{t})\triangleq\left[e^{j\frac{2\pi}{\lambda}\mathbf{t}_1^T\mathbf{a}_m},\ldots,e^{j\frac{2\pi}{\lambda}\mathbf{t}_N^T\mathbf{a}_m}\right]^T\in\mathbb{C}^{N\times 1},\qquad(1)$$

where λ is the carrier wavelength. Following [16], [17], the channel from the BS to the m-th LU follows Rician fading, i.e.,

$$\mathbf{h}_m(\mathbf{t})=\sqrt{\frac{\kappa_m\beta_m}{\kappa_m+1}}\,\bar{\mathbf{h}}_m(\mathbf{t})+\sqrt{\frac{\beta_m}{\kappa_m+1}}\,\tilde{\mathbf{h}}_m,\qquad(2)$$

where β_m is the large-scale fading coefficient, $\kappa_m\geq 0$ is the Rician factor, and $\tilde{\mathbf{h}}_m\sim\mathcal{CN}(\mathbf{0},\mathbf{I}_N)$ is the non-LoS (NLoS)

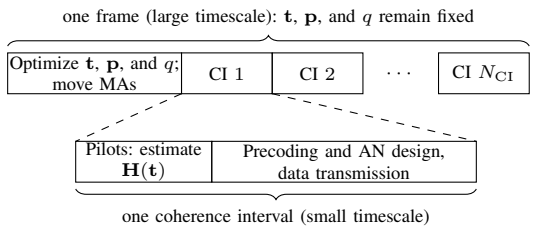


Fig. 2. Two-timescale transmission protocol (CI: coherence interval).

component.[1] Similarly, the channel from the BS to Eve is given by

$$\mathbf{g}(\mathbf{t}) = \sqrt{\frac{\kappa_e \beta_e}{\kappa_e + 1}}\, \bar{\mathbf{g}}(\mathbf{t}) + \sqrt{\frac{\beta_e}{\kappa_e + 1}}\, \tilde{\mathbf{g}}, \tag{3}$$

where β_e, κ_e, and $\tilde{\mathbf{g}} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_N)$ are defined similarly to those in (2), and $\bar{\mathbf{g}}(\mathbf{t})$ is given by (1) with $\mathbf{a}_m$ replaced by $\mathbf{a}_e \triangleq [\cos\theta_e \sin\phi_e,\ \sin\theta_e]^T$, where θ_e and ϕ_e are the elevation and azimuth AoDs of Eve, respectively. The NLoS components $\tilde{\mathbf{h}}_1, \ldots, \tilde{\mathbf{h}}_M$ and $\tilde{\mathbf{g}}$ are mutually independent. Let $\mathbf{H}(\mathbf{t}) \triangleq [\mathbf{h}_1(\mathbf{t}), \ldots, \mathbf{h}_M(\mathbf{t})] \in \mathbb{C}^{N\times M}$ denote the channel matrix from the BS to the LUs, and let $\bar{\mathbf{H}}(\mathbf{t}) \triangleq [\bar{\mathbf{h}}_1(\mathbf{t}), \ldots, \bar{\mathbf{h}}_M(\mathbf{t})]$ and $\tilde{\mathbf{H}} \triangleq [\tilde{\mathbf{h}}_1, \ldots, \tilde{\mathbf{h}}_M]$ collect its LoS and NLoS components, respectively. Moreover, $\mathbf{K} \triangleq \mathrm{diag}(\kappa_1, \ldots, \kappa_M)$ collects the Rician factors of the LUs. Note that the antenna positions affect the channels only through the LoS components $\bar{\mathbf{H}}(\mathbf{t})$ and $\bar{\mathbf{g}}(\mathbf{t})$.

B. Two-Timescale Transmission Protocol

Different from the instantaneous CSI over the entire region $\mathcal{C}$, the statistical CSI, i.e., $\{\beta_m, \kappa_m, \theta_m, \phi_m\}_{m\in\mathcal{M}}$ of the LUs and $\{\beta_e, \kappa_e, \theta_e, \phi_e\}$ of Eve, is determined by the large-scale fading and the LoS geometry, which vary much more slowly than the small-scale fading. Following [13], [15], we assume that the statistical CSI of Eve is known at the BS.[2]

Accordingly, we adopt the two-timescale protocol in Fig. 2. The time is divided into frames, each of which consists of $N_{\mathrm{CI}} \gg 1$ CIs over which the statistical CSI remains constant. At the beginning of each frame, the BS optimizes the antenna positions $\mathbf{t}$ and the long-term transmission parameters based on the statistical CSI, and moves the MAs accordingly. Then, in each CI, the BS estimates $\mathbf{H}(\mathbf{t})$ via uplink pilots by exploiting the channel reciprocity of time-division duplexing, and designs the precoding and AN. For the purpose of exposition, $\mathbf{H}(\mathbf{t})$ is assumed to be perfectly estimated, whereas $\tilde{\mathbf{g}}$ is never available at the BS. Hence, the MAs are moved only once per frame, and the CSI of the LUs is required only at the current antenna positions.

[1]As in [16], [17], the NLoS components at different antennas are assumed to be independent and identically distributed (i.i.d.), which is reasonable in rich-scattering environments with $D_{\min} \geq \lambda/2$.

[2]The statistical CSI of Eve can be obtained, e.g., from the prior knowledge of the location of Eve, by sensing its direction [15], or from its historical signals if it is an untrusted node.

C. Signal Model and Ergodic Secrecy Rate

In each CI, the BS transmits the confidential symbols $s_m \sim \mathcal{CN}(0,1)$, $m \in \mathcal{M}$, together with AN, i.e.,

$$\mathbf{x} = \sum_{m=1}^{M} \mathbf{w}_m s_m + \mathbf{z}, \tag{4}$$

where $\mathbf{w}_m \in \mathbb{C}^{N\times 1}$ is the precoding vector for the m-th LU, and $\mathbf{z} \sim \mathcal{CN}(\mathbf{0}, \mathbf{\Sigma})$ is the AN with covariance matrix $\mathbf{\Sigma} \succeq \mathbf{0}$, where $s_1, \ldots, s_M$ and $\mathbf{z}$ are mutually independent. The transmit power satisfies $\sum_{m=1}^{M} \|\mathbf{w}_m\|^2 + \mathrm{tr}(\mathbf{\Sigma}) \leq P_{\mathrm{tot}}$, where P_{tot} is the maximum transmit power of the BS. Under the two-timescale protocol, $\mathbf{w}_m$ and $\mathbf{\Sigma}$ are designed based on the instantaneous CSI of the LUs in each CI, and are thus denoted by $\mathbf{w}_m(\mathbf{H}(\mathbf{t}))$ and $\mathbf{\Sigma}(\mathbf{H}(\mathbf{t}))$ to emphasize this dependence.

The signal received at the m-th LU is $y_m = \mathbf{h}_m^H(\mathbf{t})\mathbf{x} + n_m$, where $n_m \sim \mathcal{CN}(0, \sigma_m^2)$ is the additive white Gaussian noise (AWGN). The resulting signal-to-interference-plus-noise ratio (SINR) of the m-th LU is

$$\gamma_m = \frac{|\mathbf{h}_m^H(\mathbf{t})\mathbf{w}_m|^2}{\sum_{i=1, i\neq m}^{M} |\mathbf{h}_m^H(\mathbf{t})\mathbf{w}_i|^2 + \mathbf{h}_m^H(\mathbf{t})\mathbf{\Sigma}\mathbf{h}_m(\mathbf{t}) + \sigma_m^2}. \tag{5}$$

The signal received at Eve is $y_e = \mathbf{g}^H(\mathbf{t})\mathbf{x} + n_e$ with $n_e \sim \mathcal{CN}(0, \sigma_e^2)$. Since the receiver of a passive Eve is unknown to the BS, we consider the worst case where Eve can cancel the signals of the other LUs when intercepting s_m, e.g., via multiuser detection [20], which yields a secrecy guarantee regardless of the decoding capability of Eve. The SINR of Eve for s_m is then

$$\gamma_{e,m} = \frac{|\mathbf{g}^H(\mathbf{t})\mathbf{w}_m|^2}{\mathbf{g}^H(\mathbf{t})\mathbf{\Sigma}\mathbf{g}(\mathbf{t}) + \sigma_e^2}, \tag{6}$$

which upper-bounds that of an Eve treating the signals of the other LUs as interference.

Since the instantaneous CSI of Eve is unknown, the confidential message of each LU is encoded over the CIs of a frame, which experience independent realizations of the NLoS components. By coding over these fading realizations, the following ergodic secrecy rate is achievable for the m-th LU [20], [21]:

$$R_m^{\mathrm{sec}} \triangleq \left[\bar{R}_m - \bar{R}_{e,m}\right]^+, \tag{7}$$

where

$$\bar{R}_m \triangleq \mathbb{E}\{\log_2(1+\gamma_m)\}, \quad \bar{R}_{e,m} \triangleq \mathbb{E}\{\log_2(1+\gamma_{e,m})\} \tag{8}$$

denote the ergodic rate of the m-th LU and the ergodic wiretap rate of Eve for s_m, respectively, and the expectations are taken over $\tilde{\mathbf{H}}$ and $\tilde{\mathbf{g}}$ for a given $\mathbf{t}$. The ESSR is defined as $\sum_{m=1}^{M} R_m^{\mathrm{sec}}$.

D. Problem Formulation

Let $\mathcal{W} \triangleq \{\mathbf{w}_1(\mathbf{H}(\mathbf{t})), \ldots, \mathbf{w}_M(\mathbf{H}(\mathbf{t})), \mathbf{\Sigma}(\mathbf{H}(\mathbf{t}))\}$ denote the short-timescale design policy, i.e., the mapping from the instantaneous CSI of the LUs to the precoders and the AN covariance matrix. We aim to maximize the ESSR by jointly

optimizing the long-term antenna positions and the short-timescale policy, which is formulated as

$$(\mathrm{P1}):\ \max_{\mathbf{t},\mathcal{W}}\ \sum_{m=1}^{M} R_m^{\mathrm{sec}} \tag{9a}$$

$$\text{s.t.}\ \sum_{m=1}^{M} \left\|\mathbf{w}_m(\mathbf{H}(\mathbf{t}))\right\|^2 + \mathrm{tr}\big(\mathbf{\Sigma}(\mathbf{H}(\mathbf{t}))\big) \le P_{\mathrm{tot}},$$

$$\mathbf{\Sigma}(\mathbf{H}(\mathbf{t})) \succeq \mathbf{0},\ \forall \mathbf{H}(\mathbf{t}), \tag{9b}$$

$$\mathbf{t}_n \in \mathcal{C},\ \forall n \in \mathcal{N}, \tag{9c}$$

$$\|\mathbf{t}_n - \mathbf{t}_i\| \ge D_{\min},\ \forall n,i \in \mathcal{N},\ n \ne i. \tag{9d}$$

Here, (9b) limits the transmit power in every CI, i.e., for every realization of $\mathbf{H}(\mathbf{t})$, (9c) confines the MAs to the movable region, and (9d) enforces the minimum inter-MA distance.

Problem (P1) is intractable, since the policy $\mathcal{W}$ is coupled across all the CIs through the ergodic rates in (8), which have no closed-form expressions, and since $\mathbf{t}$ is coupled with $\mathcal{W}$ under the non-convex constraint (9d). To tackle this problem, we first adopt ZF precoding and null-space AN with a long-term power allocation and derive closed-form approximations of the ergodic rates in Section III, which transform (P1) into a deterministic problem of the long-term variables. Then, we solve this problem by AO in Section IV.

III. Two-Timescale ZF-AN Design and Ergodic Secrecy Rate Analysis

A. Short-Timescale ZF Precoding and Null-Space AN

In each CI, the BS adopts ZF precoding with a long-term power allocation based on $\mathbf{H}(\mathbf{t})$, i.e., $\mathbf{w}_m(\mathbf{H}(\mathbf{t})) = \sqrt{p_m}\,\bar{\mathbf{w}}_m$, where $p_m \ge 0$ denotes the power allocated to the m-th LU, and

$$\bar{\mathbf{w}}_m \triangleq \frac{\mathbf{H}(\mathbf{t})\big(\mathbf{H}^H(\mathbf{t})\mathbf{H}(\mathbf{t})\big)^{-1}\mathbf{e}_m}{\left\|\mathbf{H}(\mathbf{t})\big(\mathbf{H}^H(\mathbf{t})\mathbf{H}(\mathbf{t})\big)^{-1}\mathbf{e}_m\right\|} \tag{10}$$

is the normalized ZF precoding vector of the m-th LU, which satisfies $\mathbf{h}_i^H(\mathbf{t})\bar{\mathbf{w}}_m = 0,\ \forall i \ne m$. Meanwhile, the AN is isotropically distributed in the null space of $\mathbf{H}^H(\mathbf{t})$ [5], [20], i.e., $\mathbf{\Sigma}(\mathbf{H}(\mathbf{t})) = q\,\mathbf{P}^{\perp}(\mathbf{t})$, where $q \ge 0$ denotes the AN power per dimension, and

$$\mathbf{P}^{\perp}(\mathbf{t}) \triangleq \mathbf{I}_N - \mathbf{H}(\mathbf{t})\big(\mathbf{H}^H(\mathbf{t})\mathbf{H}(\mathbf{t})\big)^{-1}\mathbf{H}^H(\mathbf{t}) \tag{11}$$

is the orthogonal projection onto the null space of $\mathbf{H}^H(\mathbf{t})$. Since this null space has dimension $N-M$ with probability one, we have $\mathbf{H}^H(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t}) = \mathbf{0}$ and $\mathrm{tr}(\mathbf{P}^{\perp}(\mathbf{t})) = N-M$. Hence, the AN causes no interference to the LUs, and the transmit power in each CI is $\sum_{m=1}^{M} p_m + (N-M)q$.

We refer to $\mathbf{p} \triangleq [p_1,\ldots,p_M]^T$ and q as the long-term power allocation, which is optimized based on the statistical CSI and remains fixed within a frame. In contrast, $\bar{\mathbf{w}}_m$ and $\mathbf{P}^{\perp}(\mathbf{t})$ are updated in each CI according to $\mathbf{H}(\mathbf{t})$. With this design, the SINRs in (5) and (6) reduce to

$$\gamma_m = \frac{p_m|\mathbf{h}_m^H(\mathbf{t})\bar{\mathbf{w}}_m|^2}{\sigma_m^2} = \frac{p_m}{\sigma_m^2\big[(\mathbf{H}^H(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}\big]_{m,m}}, \tag{12}$$

$$\gamma_{e,m} = \frac{p_m|\mathbf{g}^H(\mathbf{t})\bar{\mathbf{w}}_m|^2}{q\,\mathbf{g}^H(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t})\mathbf{g}(\mathbf{t}) + \sigma_e^2}, \tag{13}$$

respectively, where the second equality in (12) holds since $\mathbf{h}_m^H(\mathbf{t})\mathbf{H}(\mathbf{t})(\mathbf{H}^H(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}\mathbf{e}_m = 1$ and $\|\mathbf{H}(\mathbf{t})(\mathbf{H}^H(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}\mathbf{e}_m\|^2 = [(\mathbf{H}^H(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}]_{m,m}$.

The adopted ZF-AN design requires only the instantaneous CSI of the LUs and eliminates both the inter-user interference and the AN at the LUs. Moreover, since $\mathbf{p}$ and q depend only on the statistical CSI, no optimization problem needs to be solved in each CI. The equal power allocation in [16] and the information–AN power splitting with equal power among the LUs are special cases of this power allocation. With the ZF-AN design, (P1) is transformed into

$$(\mathrm{P2}):\ \max_{\mathbf{t},\mathbf{p},q}\ \sum_{m=1}^{M} R_m^{\mathrm{sec}}(\mathbf{t},p_m,q) \tag{14a}$$

$$\text{s.t.}\ \sum_{m=1}^{M} p_m + (N-M)q \le P_{\mathrm{tot}}, \tag{14b}$$

$$p_m \ge 0,\ \forall m \in \mathcal{M}, \quad q \ge 0, \tag{14c}$$

$$\mathbf{t}_n \in \mathcal{C},\ \forall n \in \mathcal{N}, \tag{14d}$$

$$\|\mathbf{t}_n - \mathbf{t}_i\| \ge D_{\min},\ \forall n,i \in \mathcal{N},\ n \ne i, \tag{14e}$$

where $R_m^{\mathrm{sec}}(\mathbf{t},p_m,q)$ denotes (7) with the SINRs in (12) and (13), and (14b) guarantees (9b) in every CI. Since the ergodic rates in (14a) still have no closed-form expressions, we derive their approximations in the sequel.

B. Ergodic Rate of the LUs

Define $\mathbf{B} \triangleq \mathrm{diag}\big(\frac{\beta_1}{\kappa_1+1},\ldots,\frac{\beta_M}{\kappa_M+1}\big)$ and $\mathbf{L}(\mathbf{t}) \triangleq \bar{\mathbf{H}}(\mathbf{t})\mathbf{K}^{1/2}$. From (2), the channel matrix of the LUs can be written as

$$\mathbf{H}(\mathbf{t}) = \big(\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}}\big)\mathbf{B}^{1/2}, \tag{15}$$

where $\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}}$ is the normalized channel matrix, whose LoS component is weighted by the square roots of the Rician factors. Accordingly, we define

$$\mathbf{\Phi}(\mathbf{t}) \triangleq \mathbf{L}^H(\mathbf{t})\mathbf{L}(\mathbf{t}) + N\mathbf{I}_M, \qquad \mathbf{\Omega}(\mathbf{t}) \triangleq \mathbf{\Phi}^{-1}(\mathbf{t}), \tag{16}$$

where $\mathbf{\Phi}(\mathbf{t}) = \mathbb{E}\{(\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}})^H(\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}})\}$ is the correlation matrix of the normalized channels of the LUs. The ergodic rate of the LUs under ZF precoding has been characterized in [16], [19], which is restated below for completeness.

Lemma 1 ([16], [19]): For given $\mathbf{t}$ and p_m, the ergodic rate of the m-th LU is approximated by

$$\bar{R}_m \approx \tilde{R}_m(\mathbf{t},p_m) \triangleq \log_2\big(1 + \eta_m(\mathbf{t})\,p_m\big), \tag{17}$$

where $\eta_m(\mathbf{t}) \triangleq \zeta_m(N-M)/(N\omega_m(\mathbf{t}))$ is the effective channel gain of the m-th LU, $\zeta_m \triangleq \beta_m/((\kappa_m+1)\sigma_m^2)$, and $\omega_m(\mathbf{t}) \triangleq [\mathbf{\Omega}(\mathbf{t})]_{m,m}$.

Lemma 1 follows from Jensen's inequality and a central Wishart approximation of $(\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}})^H(\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}})$ with the same mean $\mathbf{\Phi}(\mathbf{t})$ [19]. It is equivalent to the approximation in [16, Sec. IV-A], reduces to the well-known lower bound $\log_2(1 + \zeta_m p_m(N-M))$ for Rayleigh fading [23], and is conservative in the LoS-dominant regime (see Section V-B). Moreover, the antenna positions affect $\tilde{R}_m(\mathbf{t},p_m)$

only through $\omega_m(\mathbf{t})$. Since $\mathbf{\Phi}(\mathbf{t}) \succeq N\mathbf{I}_M$ and $[\mathbf{\Omega}(\mathbf{t})]_{m,m} \geq 1/[\mathbf{\Phi}(\mathbf{t})]_{m,m} = 1/(N(\kappa_m+1))$ [22], we have

$$\zeta_m(N-M) \leq \eta_m(\mathbf{t}) \leq \zeta_m(N-M)(\kappa_m+1), \tag{18}$$

where the upper bound is achieved if $\bar{\mathbf{h}}_m^H(\mathbf{t})\bar{\mathbf{h}}_i(\mathbf{t}) = 0$ for all $i \neq m$ with $\kappa_i > 0$. Hence, $1/(N\omega_m(\mathbf{t})) \in [1, \kappa_m+1]$ is the LoS gain of the m-th LU that survives the ZF precoding, which can be improved by decorrelating the LoS components of the LUs via the antenna positions.

C. Ergodic Wiretap Rate of Eve

Different from γ_m, the SINR of Eve in (13) depends on the random ZF direction $\bar{\mathbf{w}}_m$ and the random AN subspace, both of which are determined by $\mathbf{H}(\mathbf{t})$. To handle this dependence, we first characterize the information leakage and the AN received by Eve conditioned on $\mathbf{H}(\mathbf{t})$.

Lemma 2: Conditioned on $\mathbf{H}(\mathbf{t})$, the information leakage $X_m \triangleq |\mathbf{g}^H(\mathbf{t})\bar{\mathbf{w}}_m|^2$ and the AN power $Y \triangleq \mathbf{g}^H(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t})\mathbf{g}(\mathbf{t})$ received by Eve are independent, and their conditional means and variances are given by

$$\begin{aligned} \mathbb{E}\{X_m|\mathbf{H}(\mathbf{t})\} &= \tilde{\beta}_e\big(\kappa_e x_m + 1\big), \\ \mathrm{Var}\{X_m|\mathbf{H}(\mathbf{t})\} &= \tilde{\beta}_e^2\big(2\kappa_e x_m + 1\big), \end{aligned} \tag{19}$$

$$\begin{aligned} \mathbb{E}\{Y|\mathbf{H}(\mathbf{t})\} &= \tilde{\beta}_e\big(\kappa_e y + N - M\big), \\ \mathrm{Var}\{Y|\mathbf{H}(\mathbf{t})\} &= \tilde{\beta}_e^2\big(2\kappa_e y + N - M\big), \end{aligned} \tag{20}$$

where $\tilde{\beta}_e \triangleq \beta_e/(\kappa_e+1)$, $x_m \triangleq |\bar{\mathbf{g}}^H(\mathbf{t})\bar{\mathbf{w}}_m|^2$, and $y \triangleq \bar{\mathbf{g}}^H(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})$.

Proof: Let $\mathbf{U} \triangleq [\mathbf{u}_1, \ldots, \mathbf{u}_{N-M}]$ be an orthonormal basis of the null space of $\mathbf{H}^H(\mathbf{t})$. Then, $\mathbf{P}^{\perp}(\mathbf{t}) = \mathbf{U}\mathbf{U}^H$ and $Y = \sum_{i=1}^{N-M} |\mathbf{u}_i^H\mathbf{g}(\mathbf{t})|^2$. Moreover, $\bar{\mathbf{w}}_m$ is a unit-norm vector in the column space of $\mathbf{H}(\mathbf{t})$, so that $\mathbf{Q} \triangleq [\bar{\mathbf{w}}_m, \mathbf{U}]$ has orthonormal columns. On the other hand, since $\tilde{\mathbf{g}}$ is independent of $\mathbf{H}(\mathbf{t})$, (3) yields $\mathbf{g}(\mathbf{t})|\mathbf{H}(\mathbf{t}) \sim \mathcal{CN}(\sqrt{\kappa_e\tilde{\beta}_e}\,\bar{\mathbf{g}}(\mathbf{t}), \tilde{\beta}_e\mathbf{I}_N)$. Consequently, $\mathbf{Q}^H\mathbf{g}(\mathbf{t})$ is conditionally Gaussian with covariance matrix $\tilde{\beta}_e\mathbf{Q}^H\mathbf{Q} = \tilde{\beta}_e\mathbf{I}_{N-M+1}$, i.e., its entries $\bar{\mathbf{w}}_m^H\mathbf{g}(\mathbf{t}), \mathbf{u}_1^H\mathbf{g}(\mathbf{t}), \ldots, \mathbf{u}_{N-M}^H\mathbf{g}(\mathbf{t})$ are conditionally independent. Hence, $X_m = |\bar{\mathbf{w}}_m^H\mathbf{g}(\mathbf{t})|^2$ and Y are conditionally independent. Finally, (19) and (20) follow from $\mathbb{E}\{|z|^2\} = |\mu|^2 + \sigma^2$ and $\mathrm{Var}\{|z|^2\} = \sigma^4 + 2|\mu|^2\sigma^2$ for $z \sim \mathcal{CN}(\mu, \sigma^2)$, together with $\sum_{i=1}^{N-M} |\mathbf{u}_i^H\bar{\mathbf{g}}(\mathbf{t})|^2 = y$. ∎

Lemma 2 reveals that, given the channels of the LUs, the information leakage is determined only by the component of the wiretap channel along the ZF direction, whereas the AN received by Eve is determined only by its component in the null space of $\mathbf{H}^H(\mathbf{t})$. Moreover, the conditional relative variances of X_m and Y are $(2\kappa_e x_m + 1)/(\kappa_e x_m + 1)^2$ and $(2\kappa_e y + N - M)/(\kappa_e y + N - M)^2$, respectively. Hence, X_m concentrates when the LoS leakage $\kappa_e x_m$ is strong, which is the case that dominates the ergodic wiretap rate, and Y concentrates when $N - M$ or $\kappa_e y$ is large. This motivates the approximation $\mathbb{E}\{\log_2(1 + X/Y)\} \approx \log_2(1 + \mathbb{E}\{X\}/\mathbb{E}\{Y\})$ [19, Lemma 1], which is accurate when X and Y concentrate around their means and is asymptotically exact as $\kappa_e, \kappa_m \to \infty$, $\forall m \in \mathcal{M}$. Using Lemma 2 and taking the expectation over $\tilde{\mathbf{H}}$, we obtain

$$\bar{R}_{e,m} \approx \log_2\left(1 + \frac{p_m\big(\kappa_e\mathbb{E}\{x_m\} + 1\big)}{q\big(\kappa_e\mathbb{E}\{y\} + N - M\big) + \bar{\sigma}_e^2}\right), \tag{21}$$

where $\bar{\sigma}_e^2 \triangleq \sigma_e^2/\tilde{\beta}_e = \sigma_e^2(\kappa_e+1)/\beta_e$ is the normalized noise power of Eve.

In (21), $\mathbb{E}\{x_m\}$ and $\mathbb{E}\{y\}$ are the average LoS energies of Eve along the ZF direction of the m-th LU and in the AN subspace, respectively, which have no closed-form expressions. Since $\bar{\mathbf{w}}_m$ and $\mathbf{P}^{\perp}(\mathbf{t})$ are invariant to the scaling of the columns of $\mathbf{H}(\mathbf{t})$, they depend on $\mathbf{H}(\mathbf{t})$ only through the normalized channel matrix $\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}}$ in (15). We approximate $\mathbb{E}\{x_m\}$ and $\mathbb{E}\{y\}$ by replacing the inverse of the random Gram matrix of $\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}}$ with the inverse of its mean $\mathbf{\Phi}(\mathbf{t})$, where the NLoS contribution to the AN subspace is calibrated by the identity $\mathrm{tr}(\mathbf{P}^{\perp}(\mathbf{t})) = N - M$. This leads to the following closed-form approximation.

Proposition 1: Define $\mathbf{v}(\mathbf{t}) \triangleq \mathbf{L}^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t}) = \mathbf{K}^{1/2}\bar{\mathbf{H}}^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})$ and

$$c_m(\mathbf{t}) \triangleq \frac{\big|[\mathbf{\Omega}(\mathbf{t})\mathbf{v}(\mathbf{t})]_m\big|^2 + N[\mathbf{\Omega}^2(\mathbf{t})]_{m,m}}{\omega_m(\mathbf{t})}, \tag{22}$$

$$d(\mathbf{t}) \triangleq \frac{(N-M)\big(N - \mathbf{v}^H(\mathbf{t})\mathbf{\Omega}(\mathbf{t})\mathbf{v}(\mathbf{t})\big)}{N - M + N\mathrm{tr}(\mathbf{\Omega}(\mathbf{t}))}. \tag{23}$$

Then, $\mathbb{E}\{x_m\} \approx c_m(\mathbf{t})$, $\mathbb{E}\{y\} \approx d(\mathbf{t})$, and the ergodic wiretap rate of Eve for s_m is approximated by

$$\bar{R}_{e,m} \approx \tilde{R}_{e,m}(\mathbf{t}, p_m, q) \triangleq \log_2\left(1 + \frac{\xi_m(\mathbf{t})\, p_m}{q\,\psi(\mathbf{t}) + \bar{\sigma}_e^2}\right), \tag{24}$$

where $\xi_m(\mathbf{t}) \triangleq \kappa_e c_m(\mathbf{t}) + 1$ and $\psi(\mathbf{t}) \triangleq \kappa_e d(\mathbf{t}) + N - M$.

Proof: See Appendix A. ∎

The approximations $\mathbb{E}\{x_m\} \approx c_m(\mathbf{t})$ and $\mathbb{E}\{y\} \approx d(\mathbf{t})$ in Proposition 1 are exact for Rayleigh fading and asymptotically exact as the LoS components dominate, and $c_m(\mathbf{t}), d(\mathbf{t}) \in [0, N]$ hold as for their exact counterparts (see Appendix A). Accordingly, $\xi_m(\mathbf{t})$ and $\psi(\mathbf{t})$ represent the normalized information leakage for the m-th LU and the normalized AN power received by Eve, respectively. Different from the existing approximations for ZF precoding, which only involve the diagonal entries of $\mathbf{\Omega}(\mathbf{t})$, Proposition 1 captures the correlation between the LoS components of Eve and the LUs via $\mathbf{v}(\mathbf{t})$.

D. Approximate ESSR

Based on Lemma 1 and Proposition 1, the ergodic secrecy rate of the m-th LU is approximated by

$$R_m^{\mathrm{sec}} \approx \tilde{R}_m^{\mathrm{sec}}(\mathbf{t}, p_m, q) \triangleq \big[\tilde{R}_m(\mathbf{t}, p_m) - \tilde{R}_{e,m}(\mathbf{t}, p_m, q)\big]^+, \tag{25}$$

which depends only on the statistical CSI and is thus suitable for the large-timescale design. The approximate ESSR $\sum_{m=1}^{M} \tilde{R}_m^{\mathrm{sec}}(\mathbf{t}, p_m, q)$ reveals how the antenna positions affect the secrecy performance, as discussed below.

1) Secrecy mechanisms of the MAs: From (17), (22), and (23), the antenna positions affect the approximate ESSR via $\omega_m(\mathbf{t})$, $c_m(\mathbf{t})$, and $d(\mathbf{t})$, i.e., via the LoS array correlations $[\bar{\mathbf{H}}^H(\mathbf{t})\bar{\mathbf{H}}(\mathbf{t})]_{i,j} = \sum_{n=1}^{N} e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^T(\mathbf{a}_j - \mathbf{a}_i)}$ among the LUs and

$[\bar{\mathbf{H}}^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})]_i = \sum_{n=1}^{N} e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^T(\mathbf{a}_e-\mathbf{a}_i)}$ between Eve and the LUs. A smaller $\omega_m(\mathbf{t})$ yields a larger ZF gain of the m-th LU, which is the only mechanism exploited by the rate-oriented design [16]. In secure transmission, a smaller $c_m(\mathbf{t})$ and a larger $d(\mathbf{t})$ provide two additional mechanisms, i.e., less information leakage and more AN received by Eve. In particular, if $\mathbf{v}(\mathbf{t}) = \mathbf{0}$, i.e., the LoS component of Eve is orthogonal to those of all the LUs, then $c_m(\mathbf{t})$ reduces to its minimum $N[\mathbf{\Omega}^2(\mathbf{t})]_{m,m}/\omega_m(\mathbf{t})$ caused by the NLoS components, and $d(\mathbf{t})$ achieves its maximum $N(N-M)/(N-M+N\mathrm{tr}(\mathbf{\Omega}(\mathbf{t})))$ for a given $\mathbf{\Phi}(\mathbf{t})$. Hence, by decorrelating the LoS component of Eve from those of the LUs, the MAs simultaneously reduce the information leakage and increase the AN received by Eve, which is difficult for an FPA array when Eve is located in a direction close to that of an LU. Note that if $\kappa_e = 0$, then $\tilde{R}_{e,m} = \log_2(1 + p_m/(q(N-M)+\bar{\sigma}_e^2))$ does not depend on $\mathbf{t}$, i.e., the statistical CSI of Eve carries no spatial information, and only the first mechanism remains.

2) Single-LU case: If $M = 1$, then $\mathbf{\Phi}(\mathbf{t}) = N(\kappa_1+1)$ is a scalar, and $\eta_1 = \zeta_1(N-1)(\kappa_1+1)$ does not depend on $\mathbf{t}$.[3] In contrast,

$$c_1(\mathbf{t}) = \frac{\kappa_1|\bar{\mathbf{h}}_1^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2/N+1}{\kappa_1+1}, \tag{26}$$

$$d(\mathbf{t}) = \frac{(N-1)(\kappa_1+1)}{(N-1)(\kappa_1+1)+1}\left(N - \frac{\kappa_1|\bar{\mathbf{h}}_1^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2}{N(\kappa_1+1)}\right) \tag{27}$$

depend on $\mathbf{t}$ through the LoS correlation $|\bar{\mathbf{h}}_1^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2$ between the LU and Eve. Hence, different from the rate-oriented two-timescale design, where the MAs bring no gain for a single user [16], the MAs enhance the secrecy of a single LU by reducing this correlation.

3) High-SNR behavior and AN saving: Consider the information–AN power splitting with equal power among the LUs, i.e., $p_m = \alpha P_{\mathrm{tot}}/M$ and $q = (1-\alpha)P_{\mathrm{tot}}/(N-M)$, where $\alpha \in (0,1)$ is the power-splitting factor. As $P_{\mathrm{tot}} \to \infty$, we have

$$\tilde{R}_m - \tilde{R}_{e,m} = \log_2\frac{\alpha P_{\mathrm{tot}}\eta_m(\mathbf{t})}{M} - \log_2\left(1+\frac{\alpha\Theta_m(\mathbf{t})}{1-\alpha}\right) + o(1), \tag{28}$$

where $\Theta_m(\mathbf{t}) \triangleq \frac{(N-M)\xi_m(\mathbf{t})}{M\psi(\mathbf{t})}$ is the ratio of the normalized information leakage to the normalized AN power received by Eve. Hence, the high-SNR slope of the ESSR equals M regardless of $\mathbf{t}$, and the antenna positions affect the ESSR through the constant term. Moreover, the value of α that maximizes the sum of the constant terms in (28) is the root of $\sum_{m=1}^{M}\left[\frac{1}{\alpha} - \frac{1}{1-\alpha} - \frac{\Theta_m-1}{1+\alpha(\Theta_m-1)}\right] = 0$, which, for $\Theta_m = \Theta$, $\forall m \in \mathcal{M}$, is given by

$$\alpha^\star = \frac{1}{1+\sqrt{\Theta}}. \tag{29}$$

Since the MAs can decrease $c_m(\mathbf{t})$ and increase $d(\mathbf{t})$, they decrease Θ and thus increase $\alpha^\star$, i.e., they reduce the AN power required for secure transmission, which is referred to as the AN-saving effect.

[3]In fact, for $M = 1$, the exact ergodic rate $\mathbb{E}\{\log_2(1+p_1\|\mathbf{h}_1(\mathbf{t})\|^2/\sigma_1^2)\}$ does not depend on $\mathbf{t}$, since the distribution of $\|\mathbf{h}_1(\mathbf{t})\|^2$ depends on $\bar{\mathbf{h}}_1(\mathbf{t})$ only through $\|\bar{\mathbf{h}}_1(\mathbf{t})\|^2 = N$.

IV. Proposed Two-Timescale Optimization Algorithm

In this section, we first reformulate (P2) into a deterministic problem based on the approximate ESSR. Then, we decompose the resulting problem into two subproblems, i.e., the power allocation for given antenna positions and the antenna position optimization for a given power allocation, and solve them alternately.

A. Problem Reformulation

By replacing the ergodic secrecy rates in (P2) with their approximations in (25), we consider the following problem, which depends only on the statistical CSI:

$$\begin{aligned}(\mathrm{P3}):\ \max_{\mathbf{t},\mathbf{p},q}\ & F(\mathbf{t},\mathbf{p},q) \triangleq \sum_{m=1}^{M}\Bigg[\log_2\left(1+\eta_m(\mathbf{t})p_m\right) \\ & \qquad - \log_2\left(1+\frac{\xi_m(\mathbf{t})p_m}{q\psi(\mathbf{t})+\bar{\sigma}_e^2}\right)\Bigg] \quad (30a)\\ \text{s.t.}\ & \text{(14b)–(14e)},\end{aligned}$$

where the operator $[\cdot]^+$ in (25) is dropped without loss of optimality. Specifically, for any feasible $(\mathbf{t},\mathbf{p},q)$, setting $p_m = 0$ for every LU whose term in (30a) is negative yields another feasible solution, whose objective value equals $\sum_{m=1}^{M}\tilde{R}_m^{\mathrm{sec}}(\mathbf{t},p_m,q)$, since the m-th term depends only on p_m and q for a given $\mathbf{t}$. Hence, (P3) has the same optimal value as its counterpart with $[\cdot]^+$, and every term in (30a) is nonnegative at any optimal solution of (P3). In other words, the per-user power allocation allows the BS to switch off the LUs with non-positive secrecy rates. In contrast, under equal power allocation among the LUs, the operator $[\cdot]^+$ must be retained.

Problem (P3) is still non-convex due to the coupling between the antenna positions and the power allocation in (30a), the non-concavity of F w.r.t. $\mathbf{t}$, and the non-convex constraint (14e), which motivates the AO framework.

B. Power Allocation Optimization for Given $\mathbf{t}$

For given $\mathbf{t}$, the coefficients η_m, ξ_m, and ψ are constants, and (P3) reduces to

$$\begin{aligned}(\mathrm{P3.1}):\ \max_{\mathbf{p},q}\ & \sum_{m=1}^{M} f_m(p_m;q) \quad (31a)\\ \text{s.t.}\ & \text{(14b), (14c)},\end{aligned}$$

where

$$f_m(p;q) \triangleq \log_2(1+\eta_m p) - \log_2\left(1+\eta_{e,m}(q)\,p\right), \tag{32}$$

and $\eta_{e,m}(q) \triangleq \xi_m/(q\psi+\bar{\sigma}_e^2)$ is the effective channel gain of Eve for LU m, which decreases with q. Problem (P3.1) is non-convex in general due to the coupling between p_m and q. Nevertheless, for a given q, its optimal $\mathbf{p}$ can be obtained in

closed form, as shown below. The first-order and second-order derivatives of f_m w.r.t. p are given by

$$\frac{\partial f_m}{\partial p} = \frac{\eta_m - \eta_{e,m}}{\ln 2\,(1+\eta_m p)(1+\eta_{e,m}p)}, \tag{33}$$

$$\frac{\partial^2 f_m}{\partial p^2} = \frac{1}{\ln 2}\left[\frac{\eta_{e,m}^2}{(1+\eta_{e,m}p)^2} - \frac{\eta_m^2}{(1+\eta_m p)^2}\right], \tag{34}$$

where the argument q of $\eta_{e,m}$ is omitted. If $\eta_m \le \eta_{e,m}$, then $f_m(p;q) \le 0$ for all $p \ge 0$, and thus $p_m = 0$ is optimal. If $\eta_m > \eta_{e,m}$, then $\eta_m(1+\eta_{e,m}p) - \eta_{e,m}(1+\eta_m p) = \eta_m - \eta_{e,m} > 0$, i.e., $\frac{\eta_m}{1+\eta_m p} > \frac{\eta_{e,m}}{1+\eta_{e,m}p}$, which implies that f_m is strictly increasing and strictly concave in $p \ge 0$. Let $\mathcal{M}_{\rm a}(q) \triangleq \{m \in \mathcal{M} \,|\, \eta_m > \eta_{e,m}(q)\}$ denote the set of LUs that can achieve positive secrecy rates, and $P_{\rm I}(q) \triangleq P_{\rm tot} - (N-M)q$ the power available for the confidential signals. Then, the optimal $\mathbf{p}$ for a given q is given as follows.

Proposition 2 (Secrecy water-filling): For a given q with $P_{\rm I}(q) > 0$ and $\mathcal{M}_{\rm a}(q) \neq \emptyset$, the optimal power allocation is $p_m^\star = 0$ for $m \notin \mathcal{M}_{\rm a}(q)$ and

$$p_m^\star(\nu) = \left[\frac{2(\rho_m - 1)}{\sqrt{(\eta_m - \eta_{e,m})^2 + 4\eta_m\eta_{e,m}\rho_m} + \eta_m + \eta_{e,m}}\right]^+ \tag{35}$$

for $m \in \mathcal{M}_{\rm a}(q)$, where $\rho_m = \rho_m(\nu) \triangleq (\eta_m - \eta_{e,m})/(\nu \ln 2)$, and the water level $\nu > 0$ is the unique solution of $\sum_{m\in\mathcal{M}_{\rm a}(q)} p_m^\star(\nu) = P_{\rm I}(q)$.

Proof: Since the objective is strictly concave and increasing in $\{p_m\}_{m\in\mathcal{M}_{\rm a}(q)}$ and Slater's condition holds, the Karush–Kuhn–Tucker (KKT) conditions are necessary and sufficient for optimality, and the power constraint is active at the optimum [24]. Let $\nu > 0$ denote the dual variable associated with the power constraint. The KKT conditions yield $\partial f_m/\partial p = \nu$ if $p_m^\star > 0$ and $\partial f_m/\partial p|_{p=0} \le \nu$ otherwise. From (33), the former is equivalent to $(1+\eta_m p)(1+\eta_{e,m}p) = \rho_m(\nu)$, i.e., a quadratic equation in p, whose positive root, written in a numerically stable form, is given by (35). The latter is equivalent to $\rho_m(\nu) \le 1$, in which case (35) gives $p_m^\star = 0$. Finally, since $\sum_{m\in\mathcal{M}_{\rm a}(q)} p_m^\star(\nu)$ is continuous and strictly decreasing in ν over $(0, \max_{m\in\mathcal{M}_{\rm a}(q)}(\eta_m - \eta_{e,m})/\ln 2)$, ranging from $+\infty$ to 0, the water level is unique. ∎

Proposition 2 reveals that (35) reduces to the classic water-filling solution $[1/(\nu\ln 2) - 1/\eta_m]^+$ when $\eta_{e,m} \to 0$. Hence, (35) can be regarded as a secrecy water-filling solution, in which the LUs whose effective channel gains do not exceed that of Eve are switched off. Since $p_m^\star(\nu)$ is non-increasing in ν, the water level can be efficiently found by bisection.

Next, let $G(q) \triangleq \sum_{m\in\mathcal{M}} f_m(p_m^\star; q)$ denote the objective value achieved by the optimal $\mathbf{p}$ for a given q, where $G(q) = 0$ if $\mathcal{M}_{\rm a}(q) = \emptyset$ or $P_{\rm I}(q) = 0$. The optimal AN power per dimension is obtained by solving

$$q^\star = \arg \max_{0 \le q \le q_{\max}} \; G(q), \tag{36}$$

where $q_{\max} \triangleq P_{\rm tot}/(N-M)$. Increasing q reduces the effective channel gains of Eve at the cost of less power for the confidential signals. Since $G(q)$ is not necessarily unimodal, (36) is solved by a coarse grid search followed by a fine grid search around the best grid point, as summarized in Algorithm 1.

Algorithm 1 Power Allocation for Given $\mathbf{t}$

Input: $\{\eta_m, \xi_m\}_{m\in\mathcal{M}}$, ψ, $\bar{\sigma}_e^2$, $P_{\rm tot}$, and the grid size Q.
Output: $q^\star$ and $\mathbf{p}^\star$.
1: **for** each q in a uniform grid of Q points over $[0, q_{\max}]$ **do**
2: Compute $\eta_{e,m}(q)$ and $\mathcal{M}_{\rm a}(q)$, find ν by bisection such that $\sum_{m\in\mathcal{M}_{\rm a}(q)} p_m^\star(\nu) = P_{\rm I}(q)$ with $p_m^\star(\nu)$ in (35), and evaluate $G(q)$.
3: **end for**
4: Repeat Steps 1–3 over a uniform grid of Q points between the two neighbors of the best grid point, and return the best q and its $\mathbf{p}^\star$.

C. Antenna Position Optimization for Given $\mathbf{p}$ and q

For given $\mathbf{p}$ and q, (P3) reduces to

$$(\text{P3.2}): \max_{\mathbf{t}} \; F(\mathbf{t}) \tag{37a}$$

$$\text{s.t.} \;\; \mathbf{t}_n \in \mathcal{C}, \; \forall n \in \mathcal{N}, \tag{37b}$$

$$\|\mathbf{t}_n - \mathbf{t}_i\| \ge D_{\min}, \; \forall n, i \in \mathcal{N}, \; n \neq i, \tag{37c}$$

where $F(\mathbf{t}) \triangleq F(\mathbf{t}, \mathbf{p}, q)$. Problem (P3.2) is challenging for two reasons. First, according to Section III-D, $F(\mathbf{t})$ depends on $\mathbf{t}_n$ through the phases $\frac{2\pi}{\lambda}\mathbf{t}_n^T(\mathbf{a}_j - \mathbf{a}_i)$, $i, j \in \mathcal{M}\cup\{e\}$, which rotate by 2π when $\mathbf{t}_n$ moves by $\lambda/\|\mathbf{a}_j - \mathbf{a}_i\|$ along $\mathbf{a}_j - \mathbf{a}_i$. Hence, $F(\mathbf{t})$ is highly oscillatory and has numerous local maxima within $\mathcal{C}$. Second, the minimum-distance constraints are non-convex. Similar to [16], we optimize the antenna positions in an element-wise manner, i.e., we update $\mathbf{t}_n$ with the other positions fixed, for $n = 1, \ldots, N$ in turn. Let $F(\mathbf{t}_n)$ denote F as a function of $\mathbf{t}_n$, with the other positions fixed at their latest values. The subproblem for the n-th MA is given by

$$(\text{P3.2}.n): \max_{\mathbf{t}_n} \; F(\mathbf{t}_n) \tag{38a}$$

$$\text{s.t.} \;\; \mathbf{t}_n \in \mathcal{C}, \tag{38b}$$

$$\|\mathbf{t}_n - \mathbf{t}_i\| \ge D_{\min}, \; \forall i \in \mathcal{N} \setminus \{n\}. \tag{38c}$$

Although (P3.2.n) has only two real variables, $F(\mathbf{t}_n)$ is multimodal. Therefore, a purely local method, e.g., the MM method adopted in [16], converges to the local maximum closest to the current position of the MA, so that the MAs move slowly across $\mathcal{C}$ and the AO iterations may stagnate at an inferior solution (see Section V-C). To overcome this issue, we solve (P3.2.n) by a coarse-to-fine approach: in Stage 1, a global search over a grid of candidate positions locates the best basin of $F(\mathbf{t}_n)$; in Stage 2, the MM method refines the position within this basin. The global search is affordable, since $F(\mathbf{t}_n)$ can be evaluated at any candidate position with only $\mathcal{O}(M^2)$ operations by exploiting the following rank-one structure.

1) Rank-one structure: Let $\mathbf{l}_n(\mathbf{t}_n) \in \mathbb{C}^{M\times 1}$ denote the conjugate transpose of the n-th row of $\mathbf{L}(\mathbf{t})$, i.e.,

$$\mathbf{l}_n(\mathbf{t}_n) = \mathbf{K}^{1/2}\left[e^{-j\frac{2\pi}{\lambda}\mathbf{t}_n^T\mathbf{a}_1}, \ldots, e^{-j\frac{2\pi}{\lambda}\mathbf{t}_n^T\mathbf{a}_M}\right]^T. \tag{39}$$

Since $\mathbf{L}^H\mathbf{L} = \sum_{i=1}^{N} \mathbf{l}_i\mathbf{l}_i^H$ and $\mathbf{L}^H\bar{\mathbf{g}} = \sum_{i=1}^{N} \mathbf{l}_i e^{j\frac{2\pi}{\lambda}\mathbf{t}_i^T\mathbf{a}_e}$, we have

$$\mathbf{\Phi}(\mathbf{t}) = \mathbf{\Gamma}_n + \mathbf{l}_n(\mathbf{t}_n)\mathbf{l}_n^H(\mathbf{t}_n), \tag{40}$$

$$\mathbf{v}(\mathbf{t}) = \mathbf{r}_n + \mathbf{l}_n(\mathbf{t}_n)e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^T\mathbf{a}_e}, \tag{41}$$

where $\mathbf{\Gamma}_n \triangleq \sum_{i\neq n} \mathbf{l}_i(\mathbf{t}_i)\mathbf{l}_i^H(\mathbf{t}_i) + N\mathbf{I}_M$ and $\mathbf{r}_n \triangleq \sum_{i\neq n} \mathbf{l}_i(\mathbf{t}_i)e^{j\frac{2\pi}{\lambda}\mathbf{t}_i^T\mathbf{a}_e}$ do not depend on $\mathbf{t}_n$. By the Sherman–Morrison formula [22],

$$\mathbf{\Omega}(\mathbf{t}) = \mathbf{\Gamma}_n^{-1} - \frac{\mathbf{\Gamma}_n^{-1}\mathbf{l}_n(\mathbf{t}_n)\mathbf{l}_n^H(\mathbf{t}_n)\mathbf{\Gamma}_n^{-1}}{1 + \mathbf{l}_n^H(\mathbf{t}_n)\mathbf{\Gamma}_n^{-1}\mathbf{l}_n(\mathbf{t}_n)}. \tag{42}$$

Hence, for given $\mathbf{\Gamma}_n^{-1}$ and $\mathbf{r}_n$, the quantities ω_m, c_m, d, and thus $F(\mathbf{t}_n)$ can be evaluated at any candidate position with only $\mathcal{O}(M^2)$ operations.

2) Stage 1 (global search): Let $\mathcal{G} \subset \mathcal{C}$ denote a uniform grid over $\mathcal{C}$ with spacing d_{g}, and $\mathcal{G}_n \triangleq \{\mathbf{t} \in \mathcal{G} \,|\, \|\mathbf{t} - \mathbf{t}_i\| \geq D_{\min},\ \forall i \neq n\}$ the set of grid points that are feasible for (P3.2.n). Based on (42), we evaluate $F(\mathbf{t}_n)$ at all the points in $\mathcal{G}_n \cup \{\mathbf{t}_n\}$, where $\mathbf{t}_n$ is the current position of the n-th MA, and obtain

$$\hat{\mathbf{t}}_n = \arg\max_{\mathbf{t}\in\mathcal{G}_n\cup\{\mathbf{t}_n\}} F(\mathbf{t}), \tag{43}$$

which requires $\mathcal{O}(|\mathcal{G}|M^2)$ operations, e.g., $|\mathcal{G}| = 1681$ candidate positions for $A = 4\lambda$ and $d_{\mathrm{g}} = \lambda/10$. Since the current position is included in the candidates, $F(\hat{\mathbf{t}}_n) \geq F(\mathbf{t}_n)$. The spacing d_{g} is chosen as a fraction of the shortest period $\lambda/\max_{i,j\in\mathcal{M}\cup\{e\}} \|\mathbf{a}_j - \mathbf{a}_i\|$ of the phases in $F(\mathbf{t}_n)$, so that the basins of $F(\mathbf{t}_n)$ are densely sampled, and $\hat{\mathbf{t}}_n$ typically lies in the basin of a globally optimal solution of (P3.2.n).

3) Stage 2 (MM refinement): Starting from $\mathbf{t}_n^{(0)} = \hat{\mathbf{t}}_n$, we refine the position by the MM method [25], where $\mathbf{t}_n^{(k)}$ denotes the position of the n-th MA in the k-th MM iteration. To construct a surrogate function, we derive the gradient of $F(\mathbf{t}_n)$. For $u \in \{x_n, y_n\}$, let $\mathbf{D}_u \triangleq \mathrm{diag}(a_{1,u}, \ldots, a_{M,u})$, where $a_{m,x} \triangleq \cos\theta_m \sin\phi_m$ and $a_{m,y} \triangleq \sin\theta_m$ are the entries of $\mathbf{a}_m$, and let $a_{e,u}$ denote the corresponding entry of $\mathbf{a}_e$. From (39)–(41), the derivatives of $\mathbf{l}_n$, $\mathbf{v}$, and $\mathbf{\Omega}$ w.r.t. u are given by

$$\dot{\mathbf{l}}_n \triangleq \frac{\partial \mathbf{l}_n}{\partial u} = -j\frac{2\pi}{\lambda}\mathbf{D}_u\mathbf{l}_n, \tag{44}$$

$$\dot{\mathbf{v}} \triangleq \frac{\partial \mathbf{v}}{\partial u} = j\frac{2\pi}{\lambda}\big(a_{e,u}\mathbf{I}_M - \mathbf{D}_u\big)\mathbf{l}_n e^{j\frac{2\pi}{\lambda}\mathbf{t}_n^T\mathbf{a}_e}, \tag{45}$$

$$\dot{\mathbf{\Omega}} \triangleq \frac{\partial \mathbf{\Omega}}{\partial u} = -\mathbf{\Omega}\big(\dot{\mathbf{l}}_n\mathbf{l}_n^H + \mathbf{l}_n\dot{\mathbf{l}}_n^H\big)\mathbf{\Omega}, \tag{46}$$

where (46) follows from $\partial\mathbf{\Phi}^{-1} = -\mathbf{\Phi}^{-1}(\partial\mathbf{\Phi})\mathbf{\Phi}^{-1}$. Based on (44)–(46), the derivatives $\dot{\omega}_m \triangleq \partial\omega_m/\partial u$, $\dot{c}_m \triangleq \partial c_m/\partial u$, and $\dot{d} \triangleq \partial d/\partial u$ are derived in closed form in Appendix B. Let $\chi_m \triangleq \xi_m p_m$ and $\Upsilon \triangleq q\psi + \bar{\sigma}_e^2$, whose derivatives are $\dot{\chi}_m = \kappa_e p_m \dot{c}_m$ and $\dot{\Upsilon} = \kappa_e q\dot{d}$, respectively. Moreover, $\dot{\eta}_m = -\eta_m\dot{\omega}_m/\omega_m$. By the chain rule,

$$\frac{\partial F}{\partial u} = \frac{1}{\ln 2}\sum_{m=1}^{M}\left[\frac{p_m\dot{\eta}_m}{1+\eta_m p_m} - \frac{\kappa_e p_m\dot{c}_m}{\chi_m + \Upsilon} + \frac{\kappa_e q\chi_m\dot{d}}{\Upsilon(\chi_m+\Upsilon)}\right], \tag{47}$$

and the gradient of F w.r.t. $\mathbf{t}_n$ is $\nabla_{\mathbf{t}_n}F = [\partial F/\partial x_n,\ \partial F/\partial y_n]^T$.

Since $F(\mathbf{t}_n)$ is twice continuously differentiable and $\mathcal{C}$ is compact, $\nabla_{\mathbf{t}_n}F$ is Lipschitz continuous over $\mathcal{C}$ with some constant $\mathcal{L}_n$. According to the descent lemma [26, Lemma 5.7], for any $\delta_n \geq \mathcal{L}_n$, the following quadratic function is a lower bound of $F(\mathbf{t}_n)$, which is tight at $\mathbf{t}_n^{(k)}$:

$$F(\mathbf{t}_n) \geq \hat{F}_n(\mathbf{t}_n|\mathbf{t}_n^{(k)}) \triangleq F(\mathbf{t}_n^{(k)}) + \nabla_{\mathbf{t}_n}F(\mathbf{t}_n^{(k)})^T\big(\mathbf{t}_n - \mathbf{t}_n^{(k)}\big) - \frac{\delta_n}{2}\big\|\mathbf{t}_n - \mathbf{t}_n^{(k)}\big\|^2. \tag{48}$$

Since $\mathcal{L}_n$ is difficult to obtain, δ_n is determined by backtracking [26], i.e., starting from $\max\{\delta_0, \delta_n^{\mathrm{pre}}/2\}$, where $\delta_0 > 0$ is a constant and δ_n^{pre} is the previously accepted value, δ_n is doubled until the update below satisfies $F(\mathbf{t}_n^{(k+1)}) \geq \hat{F}_n(\mathbf{t}_n^{(k+1)}|\mathbf{t}_n^{(k)})$, which holds once $\delta_n \geq \mathcal{L}_n$.

To handle the minimum-distance constraints, we note that $\|\mathbf{t}_n - \mathbf{t}_i\|^2$ is convex w.r.t. $\mathbf{t}_n$ and is thus lower-bounded by its first-order Taylor expansion at $\mathbf{t}_n^{(k)}$ [16], i.e.,

$$\|\mathbf{t}_n - \mathbf{t}_i\|^2 \geq \|\mathbf{\Delta}_{n,i}^{(k)}\|^2 + 2\big(\mathbf{\Delta}_{n,i}^{(k)}\big)^T\big(\mathbf{t}_n - \mathbf{t}_n^{(k)}\big), \tag{49}$$

where $\mathbf{\Delta}_{n,i}^{(k)} \triangleq \mathbf{t}_n^{(k)} - \mathbf{t}_i$. Replacing $\|\mathbf{t}_n - \mathbf{t}_i\|^2$ in the minimum-distance constraints with the right-hand side of (49) yields the linear constraint

$$\big(\mathbf{\Delta}_{n,i}^{(k)}\big)^T\mathbf{t}_n \geq \big(\mathbf{\Delta}_{n,i}^{(k)}\big)^T\mathbf{t}_n^{(k)} + \frac{D_{\min}^2 - \|\mathbf{\Delta}_{n,i}^{(k)}\|^2}{2}, \quad \forall i \neq n, \tag{50}$$

whose feasible set is a convex subset of that of the minimum-distance constraints w.r.t. $\mathbf{t}_n$. Based on (48) and (50), $\mathbf{t}_n$ is updated by solving

$$\mathbf{t}_n^{(k+1)} = \arg\max_{\mathbf{t}_n}\ \hat{F}_n(\mathbf{t}_n|\mathbf{t}_n^{(k)}) \quad \text{s.t. } \mathbf{t}_n \in \mathcal{C},\ (50). \tag{51}$$

By completing the square, (51) is equivalent to projecting $\mathbf{t}_n^{(k)} + \nabla_{\mathbf{t}_n}F(\mathbf{t}_n^{(k)})/\delta_n$ onto the convex polygon defined by $\mathcal{C}$ and (50). Its optimal solution lies at the projected point itself (if feasible), on an edge, or at a vertex of the polygon, and can thus be obtained exactly by enumerating the projections onto the $N+3$ constraint lines and their pairwise intersections. Since $\mathbf{t}_n^{(k)}$ is feasible for (51), the update satisfies

$$F(\mathbf{t}_n^{(k+1)}) \geq \hat{F}_n(\mathbf{t}_n^{(k+1)}|\mathbf{t}_n^{(k)}) \geq \hat{F}_n(\mathbf{t}_n^{(k)}|\mathbf{t}_n^{(k)}) = F(\mathbf{t}_n^{(k)}), \tag{52}$$

and $\mathbf{t}_n^{(k+1)}$ is feasible for (P3.2.n) due to (49). The MM iterations are repeated until the fractional increase of F is below ϵ_{in} or I_{in} iterations are reached.

D. Overall Algorithm, Convergence, and Complexity

The overall algorithm is summarized in Algorithm 2, whose convergence, initialization, and complexity are discussed below.

Convergence: In each iteration of Algorithm 2, neither the global search in (43) nor the MM updates in (52) decrease the objective value of (P3), and the power allocation is updated only if the objective value does not decrease. Hence, the objective value is non-decreasing over the iterations. Moreover, it is upper-bounded, e.g., $F \leq \sum_{m=1}^{M}\log_2(1+\zeta_m(N-M)(\kappa_m+1)P_{\mathrm{tot}})$ according to (18) and $p_m \leq P_{\mathrm{tot}}$. Therefore, the objective value of Algorithm 2 is guaranteed to converge.

Algorithm 2 Two-Timescale AO Algorithm for (P3)

Input: Statistical CSI, $P_{\rm tot}$, $\mathcal{C}$, $D_{\min}$, $d_{\rm g}$, δ_0, $\epsilon_{\rm in}$, $I_{\rm in}$, ϵ, and $I_{\max}$.
Output: $\mathbf{t}^\star$, $\mathbf{p}^\star$, and $q^\star$.

1: Initialize a feasible $\mathbf{t}^{(0)}$, obtain $(\mathbf{p}^{(0)}, q^{(0)})$ by Algorithm 1, and set $\ell = 0$.
2: **repeat**
3: **for** $n = 1$ to N **do**
4: *Stage 1:* Compute $\mathbf{\Gamma}_n^{-1}$ and $\mathbf{r}_n$, and obtain $\hat{\mathbf{t}}_n$ by (43).
5: *Stage 2:* Starting from $\mathbf{t}_n^{(0)} = \hat{\mathbf{t}}_n$, update $\mathbf{t}_n$ by (51) with backtracking on δ_n until the fractional increase of F is below $\epsilon_{\rm in}$ or $I_{\rm in}$ iterations are reached.
6: **end for**
7: Obtain $(\mathbf{p}', q')$ by Algorithm 1 for the updated positions $\mathbf{t}^{(\ell+1)}$. Set $(\mathbf{p}^{(\ell+1)}, q^{(\ell+1)}) = (\mathbf{p}', q')$ if this does not decrease F, and $(\mathbf{p}^{(\ell+1)}, q^{(\ell+1)}) = (\mathbf{p}^{(\ell)}, q^{(\ell)})$ otherwise.
8: $\ell \leftarrow \ell + 1$.
9: **until** the fractional increase of F is below ϵ or $\ell = I_{\max}$

Initialization: Since (P3) is non-convex, the solution obtained by Algorithm 2 may depend on the initial positions. Owing to the global search in Stage 1, each MA can move to any feasible grid point in $\mathcal{C}$ in every iteration, which reduces this dependence compared with purely local updates. To further improve the solution, Algorithm 2 can be executed from several initializations, e.g., the FPA array and random feasible positions, and the solution with the largest objective value is selected.

Complexity: In each iteration, forming $\mathbf{\Phi}$ and $\mathbf{\Omega}$ requires $\mathcal{O}(NM^2 + M^3)$ operations. For each MA, $\mathbf{\Gamma}_n^{-1}$ is obtained from $\mathbf{\Omega}$ by a rank-one update. Stage 1 requires $\mathcal{O}(|\mathcal{G}|(M^2 + N))$ operations, including the feasibility check of the grid points, where $|\mathcal{G}| = (\lfloor A/d_{\rm g} \rfloor + 1)^2$. In Stage 2, the gradient in (47) and each evaluation of F require $\mathcal{O}(M^2)$ operations based on (42), and solving (51) by enumeration requires $\mathcal{O}(N^3)$ operations. Algorithm 1 requires $\mathcal{O}(QI_{\rm b}M)$ operations, where $I_{\rm b}$ is the number of bisection iterations. Hence, the complexity of Algorithm 2 is $\mathcal{O}\big(I_{\rm o}\big(NM^2 + M^3 + N|\mathcal{G}|(M^2+N) + NI_{\rm in}J(M^2+N^3) + QI_{\rm b}M\big)\big)$, where $I_{\rm o}$ is the number of AO iterations and J is the average number of backtracking trials. In contrast to stochastic optimization [18], this complexity does not depend on the number of channel realizations. Moreover, since Algorithm 2 depends only on the slowly varying statistical CSI, it is executed only once per frame, whereas the ZF precoding and AN generation in each CI require only $\mathcal{O}(NM^2 + M^3)$ operations.

V. Simulation Results

In this section, we first verify the accuracy of the closed-form approximations and the convergence of Algorithm 2, and then compare the proposed design with benchmark schemes.

A. Simulation Setup and Benchmarks

Unless otherwise specified, we set $N = 8$, $M = 3$, $A = 4\lambda$, $D_{\min} = \lambda/2$, $\kappa_m = \kappa_e = 10$ dB, $\beta_m = \beta_e = 1$, and $\sigma_m^2 = \sigma_e^2 = \sigma^2$, such that $P_{\rm tot}/\sigma^2$ is the transmit SNR, which is 20 dB by default. The elevation and azimuth AoDs of the LUs are uniformly distributed within $[-\pi/3, \pi/3]$. To model a challenging scenario, Eve is located in a direction close to that of LU 1, as illustrated in Fig. 1, i.e., $\theta_e = \theta_1 + \Delta\cos\tilde{\phi}$ and $\phi_e = \phi_1 + \Delta\sin\tilde{\phi}$, where $\Delta = 0.1$ rad is the angular offset, and $\tilde{\phi}$ is uniformly distributed over $[0, 2\pi)$ to randomize the direction of the offset.

For Algorithm 2, the grid spacing in Stage 1 is $d_{\rm g} = \lambda/10$, which is less than one third of the shortest phase period in Section IV-C. The MM refinement in Stage 2 uses $\delta_0 = 1$ and stops when the fractional increase of F is below $\epsilon_{\rm in} = 10^{-6}$ or after $I_{\rm in} = 20$ iterations, and Algorithm 2 stops when the fractional increase of F is below $\epsilon = 10^{-4}$ or after $I_{\max} = 50$ iterations. Algorithm 1 uses $Q = 41$ grid points and 60 bisection iterations. Algorithm 2 is executed from three initializations, i.e., a 2×4 uniform planar array (UPA) with $\lambda/2$ spacing and two random feasible positions, and the best solution is selected. All the results are averaged over 100 random realizations of the AoDs of the LUs and Eve shared by all the schemes, which is sufficient for the comparisons, since the standard error of the average ESSR of the proposed design at 20 dB is 0.11 bps/Hz, i.e., 0.5% of its value. Since the closed-form approximations in Section III are used only for optimization, the ergodic rates in (8) of each design are estimated by averaging the instantaneous rates with the exact SINRs in (12) and (13) over 5×10^3 independent channel realizations, and the ESSR is then computed by (7).

We compare the proposed design with three benchmarks: 1) **Rate-oriented MA**: The antenna positions maximize the approximate ergodic sum rate of the LUs with equal power, i.e., $\sum_{m=1}^{M} \log_2(1 + \eta_m(\mathbf{t})P_{\rm tot}/M)$, which ignores Eve [16]. For a fair comparison, this problem is solved by the position update in Section IV-C with the same initializations. This benchmark reveals the gain of the secrecy-aware positioning enabled by Proposition 1. 2) **AS**: Similar to [15], the BS selects N of $2N$ candidate antennas, which form a UPA with 4 rows and 4 columns and $\lambda/2$ spacing, by an exhaustive search over all the 12870 possible subsets, where the selection metric is the approximate ESSR in (P3). AS requires no antenna movement but twice as many antennas and an RF switching network. 3) **FPA**: The BS adopts the 2×4 UPA with $\lambda/2$ spacing. All the schemes adopt the power allocation in Algorithm 1, so that their performance differences are caused only by the antenna positions.

B. Accuracy of the Closed-Form Approximations

Fig. 3 evaluates the accuracy of Lemma 1 and Proposition 1 at the designs obtained by Algorithm 2, where 10^4 channel realizations are used for the Monte Carlo simulations. Since these designs are optimized based on the closed-form approximations, Fig. 3 verifies whether the approximations remain accurate at the operating points selected by the optimization. It is observed that Proposition 1 closely matches the Monte Carlo results over the entire ranges of the SNR and the Rician factor, with a maximum absolute error of 0.05 bps/Hz. In contrast, Lemma 1 underestimates the ergodic sum rate of the LUs, and the gap increases with the Rician factor and saturates at high SNR. This is because, in addition to Jensen's inequality, the central Wishart approximation in Lemma 1 is exact for Rayleigh fading but overestimates $\mathbb{E}\{[(\mathbf{H}^H(\mathbf{t})\mathbf{H}(\mathbf{t}))^{-1}]_{m,m}\}$

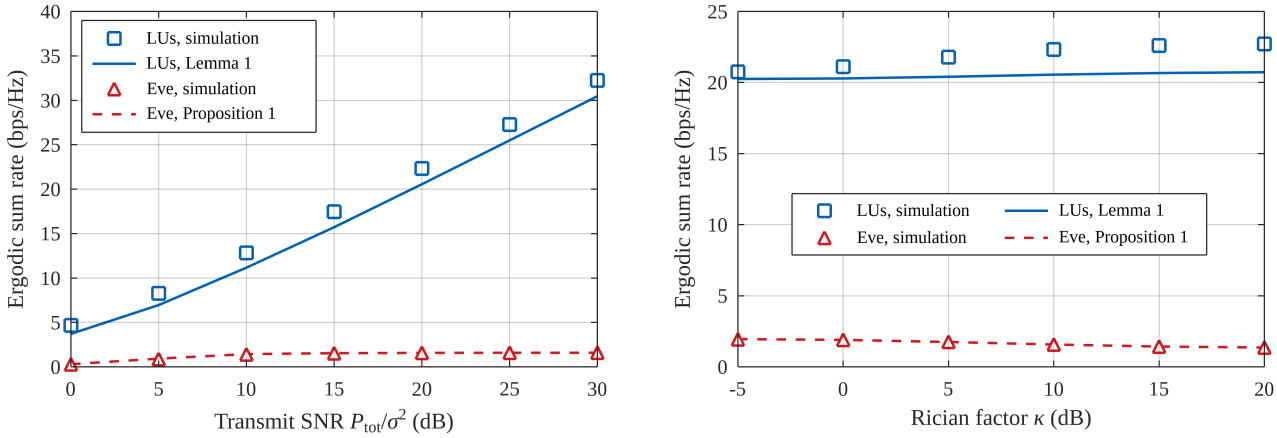


Fig. 3. Ergodic sum rate of the LUs, $\sum_{m=1}^{M} \bar{R}_m$, and ergodic wiretap sum rate of Eve, $\sum_{m=1}^{M} \bar{R}_{e,m}$, achieved by the proposed design, obtained by Monte Carlo simulations (markers) and by the closed-form approximations in Lemma 1 and Proposition 1 (lines): (a) versus the transmit SNR P_{tot}/σ^2 with $\kappa_m = \kappa_e = 10$ dB; (b) versus the Rician factor $\kappa = \kappa_m = \kappa_e$ with $P_{\text{tot}}/\sigma^2 = 20$ dB.

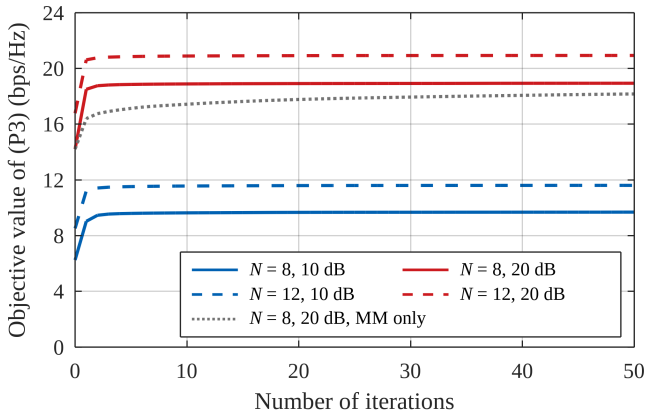


Fig. 4. Convergence behaviour of Algorithm 2 for different numbers of MAs N and transmit SNRs P_{tot}/σ^2, averaged over 100 realizations of the AoDs (UPA initialization). "MM only": Algorithm 2 without the global search in Stage 1.

by a factor approaching $N/(N-M)$ as the LoS components dominate, which leads to a gap of up to about $M\log_2(N/(N-M)) \approx 2$ bps/Hz in the sum rate for $N = 8$ and $M = 3$. Hence, the approximate ESSR in (25) is a conservative estimate of the actual ESSR. Nevertheless, at each SNR in Fig. 5(a), this gap differs by less than 0.3 bps/Hz among the four schemes, which is smaller than the ESSR gains of the proposed design. Hence, the approximation preserves the advantage of the proposed design over the benchmarks and is suitable as the design objective.

C. Convergence Behaviour of Algorithm 2

Fig. 4 shows the objective value of (P3) versus the number of iterations of Algorithm 2 for $N \in \{8, 12\}$ and $P_{\text{tot}}/\sigma^2 \in \{10, 20\}$ dB, where the UPA initialization is adopted (3×4 UPA for $N = 12$). To show the complete trajectories, Algorithm 2 is run for 50 iterations without the stopping criterion. It is observed that, for all the settings, the objective value increases monotonically and converges within about 15 iterations, where most of the improvement is achieved in the first few iterations. This is because the global search in Stage 1 moves each MA directly to the best basin of $F(\mathbf{t}_n)$, so that the positions do not need to traverse the movable region through local updates. In contrast, with only the MM updates, the objective value increases much more slowly and remains lower after 50 iterations, since the MAs are trapped in the local maxima near their initial positions.

D. ESSR and AN Power Versus Transmit SNR

Fig. 5(a) shows that the proposed design achieves the highest ESSR over the entire SNR range. At $P_{\text{tot}}/\sigma^2 = 20$ dB, it improves the ESSR by 14.8%, 20.4%, and 31.2% compared with the rate-oriented MA, AS, and the FPA, respectively. Although the rate-oriented MA exploits the same movable region, its gain over the FPA is less than half of that of the proposed design. This is because the rate-oriented positions only decorrelate the LUs, whereas the proposed positions additionally reduce the information leakage and increase the AN received by Eve (see Section III-D). Moreover, AS improves the ESSR over the FPA by only 9.0%, since its $2N$ candidates form a compact UPA with an aperture of only $1.5\lambda \times 1.5\lambda$, which provides limited spatial resolution for decorrelating the LoS components of Eve and LU 1. As the SNR increases, the ESSR gains of the proposed design approach constants, which is consistent with Section III-D: the antenna positions improve the constant term rather than the slope of the ESSR at high SNR.

Fig. 5(b) shows that the optimized AN power fraction increases with the SNR and saturates at high SNR, which is consistent with (29). Moreover, the proposed design requires considerably less AN power than all the benchmarks, whereas the rate-oriented MA saves little AN power compared with the FPA. This verifies the AN-saving effect in Section III-D:

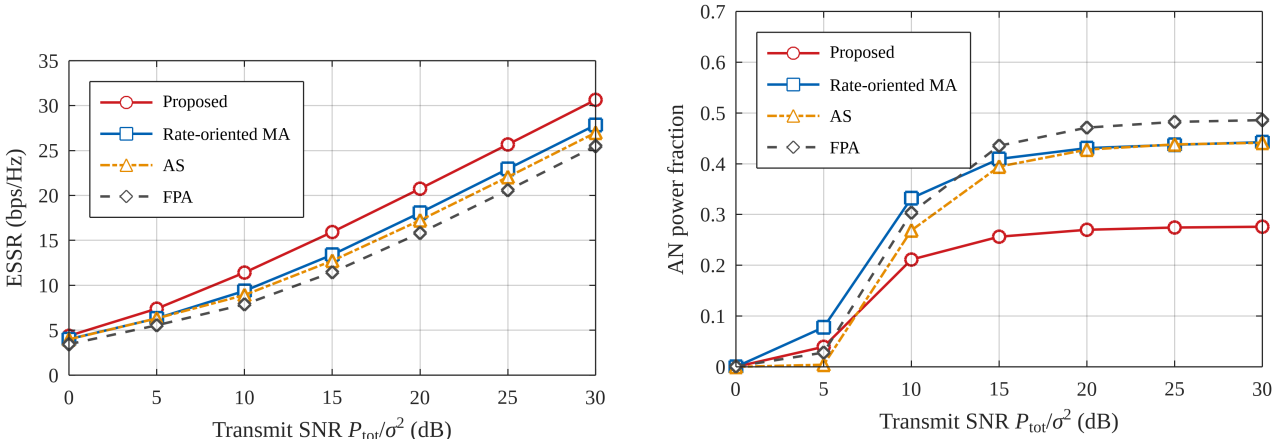


Fig. 5. Performance versus the transmit SNR P_{tot}/σ^2: (a) ESSR; (b) optimized AN power fraction $(N-M)q/P_{\text{tot}}$.

TABLE I
ESSR (BPS/HZ) UNDER DIFFERENT EAVESDROPPER MODELS

	Worst-case Eve (6)			TIN Eve (53)		
P_{tot}/σ^2 (dB)	10	20	30	10	20	30
Proposed	**11.42**	**20.75**	**30.64**	**11.85**	**21.24**	**31.14**
Rate-oriented MA	9.38	18.08	27.84	10.53	19.55	29.38
AS	8.92	17.24	26.98	9.77	18.58	28.39
FPA	7.88	15.82	25.49	8.70	17.24	27.00
Proposed w/o AN	9.94	13.06	13.85	11.00	18.70	27.14

TABLE II
ESSR (BPS/HZ) VERSUS THE AOD ERROR ε_{e} OF EVE ($P_{\text{tot}}/\sigma^2 = 20$ DB)

ε_{e} (rad)	0	0.01	0.02	0.05	0.1
Proposed	**20.75**	**20.73**	**20.65**	**20.13**	**18.73**
Rate-oriented MA	18.08	18.07	18.07	18.02	17.93
AS	17.24	17.24	17.22	17.16	16.97
FPA	15.82	15.82	15.81	15.81	15.80

since the proposed positions reduce the information leakage and increase the AN received by Eve, more power can be allocated to the confidential signals.

E. Robustness to the Eavesdropper Model and the AoD Error of Eve

All the designs are optimized under the worst-case eavesdropper model in (6) with known statistical CSI of Eve. Tables I and II examine whether their performance relies on these assumptions. Table I evaluates the designs also against an Eve that treats the signals of the other LUs as interference (TIN), whose SINR is

$$\gamma_{e,m}^{\text{TIN}} = \frac{p_m|\mathbf{g}^H(\mathbf{t})\bar{\mathbf{w}}_m|^2}{\sum_{i=1,i\neq m}^{M} p_i|\mathbf{g}^H(\mathbf{t})\bar{\mathbf{w}}_i|^2 + q\,\mathbf{g}^H(\mathbf{t})\mathbf{P}^{\perp}(\mathbf{t})\mathbf{g}(\mathbf{t}) + \sigma_e^2}. \tag{53}$$

Two observations can be made. First, the proposed design achieves the highest ESSR under both models, which confirms that its gain is not an artifact of the worst-case assumption. Moreover, its ESSR under the worst-case model is only slightly lower than that against the TIN Eve, i.e., the secrecy guarantee of the worst-case model comes at a marginal cost. Second, for the proposed design without AN, i.e., Algorithm 2 with $q = 0$, the ESSR increases with the SNR against the TIN Eve, for which the inter-user interference acts as AN, but saturates under the worst-case model [4]. Hence, the AN is indispensable for a secrecy performance that does not rely on the decoding capability of Eve.

Table II considers imperfect statistical CSI of Eve, where all the schemes are designed with an AoD of Eve that deviates from the true one by ε_{e} in a random direction, and are evaluated with the true AoD. It is observed that the ESSR of the proposed design decreases only slightly for $\varepsilon_{\text{e}} \leq 0.05$ rad, and the proposed design still outperforms all the benchmarks for $\varepsilon_{\text{e}} = \Delta = 0.1$ rad, i.e., an error as large as the angular offset between Eve and LU 1. This is because the LoS correlations between Eve and the LUs vary smoothly with the AoD of Eve, so that the positions optimized for a moderately erroneous AoD still decorrelate the actual LoS components.

F. Impact of the Region Size and the Angular Offset

Fig. 6(a) shows the ESSR versus the side length A of the movable region, where we also consider AS with region-wide candidates, i.e., a 4×4 candidate UPA spanning $\mathcal{C}$ with spacing $A/3$. Note that the FPA and AS with the compact candidate array do not depend on A. It is observed that the gain of the proposed design over the FPA increases from 9.2% to 38.9% as A increases from 1.5λ to 6λ. This is because a larger region provides more spatial degrees of freedom for decorrelating the LoS components of Eve and the LUs. In contrast, the ESSR of the rate-oriented MA increases much more slowly with A, since it does not exploit the enlarged region to reduce the information leakage. AS with region-wide candidates also benefits from the enlarged aperture and the Eve-aware selection metric, but the proposed design still outperforms it for $A \geq 2\lambda$ with half the number of antennas, since the MAs can be placed at arbitrary positions in $\mathcal{C}$ rather than at $2N$ fixed candidate positions. For $A = 1.5\lambda$, the two AS schemes coincide and perform comparably to the proposed design, since the candidate array covers the entire region.

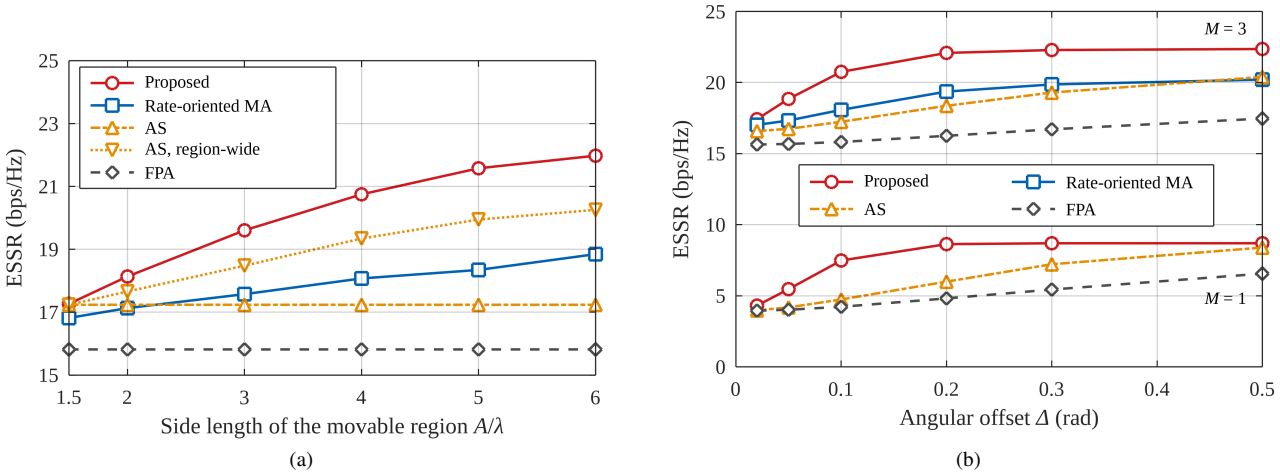


Fig. 6. ESSR versus (a) the side length A of the movable region and (b) the angular offset Δ between Eve and LU 1 for $M = 3$ and $M = 1$.

Fig. 6(b) shows the ESSR versus the angular offset Δ for $M = 3$ and $M = 1$. For $M = 3$, all the schemes benefit from a larger Δ, since Eve becomes more separable from LU 1. For $\Delta = 0.02$ rad, Eve is almost aligned with LU 1, and the proposed design outperforms the rate-oriented MA only slightly, since their LoS components can hardly be decorrelated with an aperture of 4λ. As Δ increases, the gain of the proposed design first increases and then decreases, since Eve eventually becomes separable even with Eve-unaware positions. For $M = 1$, the rate-oriented MA coincides with the FPA, since the ergodic rate of the LU does not depend on the antenna positions (see Section III-D). In contrast, the proposed design significantly improves the ergodic secrecy rate, e.g., by 77.1% over the FPA for $\Delta = 0.1$ rad, by reducing the LoS correlation $|\bar{\mathbf{h}}_1^H(\mathbf{t})\bar{\mathbf{g}}(\mathbf{t})|^2$ in (26) and (27). This confirms that, different from the rate-oriented design, the MAs enhance secrecy even for a single LU.

G. Benefit of the Long-Term Power Allocation

Fig. 7 evaluates the benefit of the proposed long-term power allocation, i.e., the secrecy water-filling in Proposition 2 combined with the optimized AN power. It is compared with two EP benchmarks with $p_m = \alpha P_{\text{tot}}/M$ and $q = (1-\alpha)P_{\text{tot}}/(N-M)$, where α is either optimized by a one-dimensional search, similar to [5], or fixed to 0.5, similar to [15]. For all the schemes, the antenna positions are optimized by Algorithm 2, where the objective of the EP benchmarks retains $[\cdot]^+$, since the LUs cannot be switched off (see Section IV-A). The large-scale fading of the LUs is $[\beta_1, \beta_2, \beta_3] = [0, -\varsigma/2, -\varsigma]$ dB, where ς is referred to as the fading spread.

Fig. 7(a) shows that, for $\varsigma = 20$ dB, the proposed power allocation outperforms both EP benchmarks. Its gain over EP with optimized α is significant at low-to-moderate SNR, e.g., 0.65 bps/Hz at 10 dB, but becomes marginal at 20 dB. This is because the proposed power allocation switches off or reduces the power of the LUs whose effective channel gains are comparable to that of Eve (Proposition 2), whereas at high SNR, all the LUs achieve positive secrecy rates and nearly equal powers become optimal, similar to conventional water-filling. Fig. 7(b) shows that EP with optimized α performs comparably to the proposed power allocation for small fading spreads, e.g., $\varsigma \leq 5$ dB, since the LUs have similar effective channel gains, which are further balanced by the optimized positions, whereas the gain of the proposed power allocation increases with ς for larger fading spreads. Moreover, EP with $\alpha = 0.5$ performs worst in all the cases, especially at low SNR, where little or no AN is needed. Hence, both components of the long-term power allocation are important: the information–AN power splitting improves the ESSR in all the cases, whereas the per-user power allocation provides additional gains for heterogeneous LUs at low-to-moderate SNR.

VI. Conclusion

This paper proposed a two-timescale secure transmission framework for an MA-enabled multiuser downlink system with a passive Eve, where the antenna positions and the long-term power allocation are optimized based on the statistical CSI, and ZF precoding and null-space AN are designed based on the instantaneous CSI of the LUs. We derived a closed-form approximation of the ergodic wiretap rate, which reveals that, beyond decorrelating the LUs, the MAs enhance secrecy by reducing the information leakage and increasing the AN received by Eve. Based on this result, we developed an AO algorithm with secrecy water-filling and a two-stage position update, which exhibits fast and monotonic convergence. Simulation results demonstrated that the proposed secrecy-aware positioning significantly outperforms the rate-oriented two-timescale MA design, AS, and FPA arrays, enhances the secrecy even for a single LU, and reduces the AN power required for secure transmission. The analysis in this paper assumes a single-antenna Eve and perfect CSI of the LUs, and extending it to multiple multi-antenna eavesdroppers and imperfect CSI is an important direction for future work.

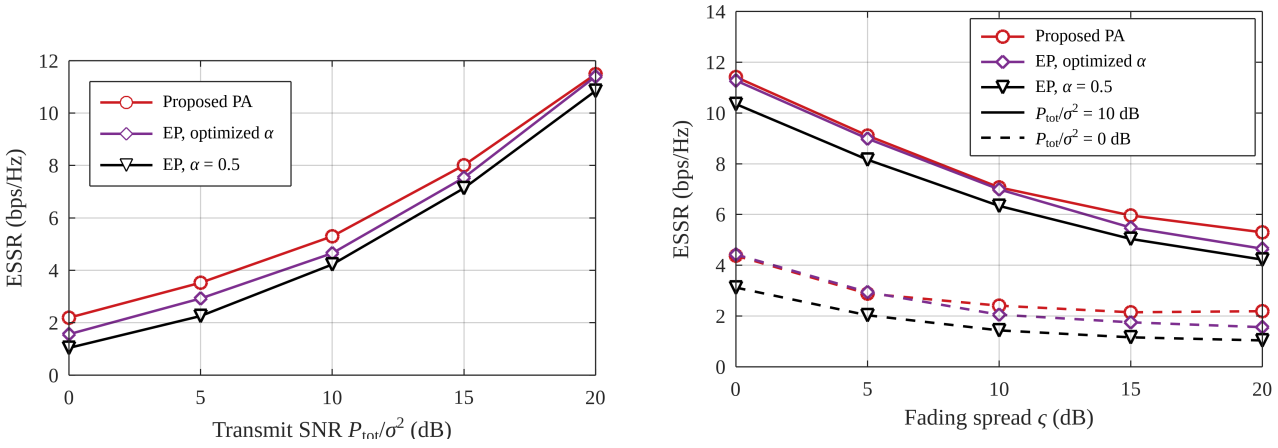


Fig. 7. ESSR of the proposed power allocation (PA) and two equal-power (EP) benchmarks versus (a) the transmit SNR with $\varsigma = 20$ dB and (b) the fading spread ς.

Appendix A
Proof of Proposition 1

We first establish the following lemma for a generic non-central Gaussian matrix, and then apply it to (21).

Lemma 3: Let $\mathbf{F} = \bar{\mathbf{F}} + \mathbf{W} \in \mathbb{C}^{N\times k}$ with $k < N$, where $\bar{\mathbf{F}}$ is deterministic and the entries of $\mathbf{W}$ are i.i.d. $\mathcal{CN}(0,1)$ random variables. Let $\mathbf{b} \in \mathbb{C}^{N\times 1}$ be a deterministic vector, and define $\mathbf{\Phi_F} \triangleq \mathbb{E}\{\mathbf{F}^H\mathbf{F}\} = \bar{\mathbf{F}}^H\bar{\mathbf{F}} + N\mathbf{I}_k$, $\bar{\mathbf{w}}_m(\mathbf{F}) \triangleq \mathbf{F}(\mathbf{F}^H\mathbf{F})^{-1}\mathbf{e}_m/\|\mathbf{F}(\mathbf{F}^H\mathbf{F})^{-1}\mathbf{e}_m\|$, and $\mathbf{P}_{\mathbf{F}}^{\perp} \triangleq \mathbf{I}_N - \mathbf{F}(\mathbf{F}^H\mathbf{F})^{-1}\mathbf{F}^H$. Then, we have

$$\mathbb{E}\{|\mathbf{b}^H\bar{\mathbf{w}}_m(\mathbf{F})|^2\} \approx \frac{\left|[\mathbf{\Phi}_{\mathbf{F}}^{-1}\bar{\mathbf{F}}^H\mathbf{b}]_m\right|^2 + \|\mathbf{b}\|^2[\mathbf{\Phi}_{\mathbf{F}}^{-2}]_{m,m}}{[\mathbf{\Phi}_{\mathbf{F}}^{-1}]_{m,m}}, \tag{54}$$

$$\mathbb{E}\{\mathbf{b}^H\mathbf{P}_{\mathbf{F}}^{\perp}\mathbf{b}\} \approx \frac{(N-k)\left(\|\mathbf{b}\|^2 - \mathbf{b}^H\bar{\mathbf{F}}\mathbf{\Phi}_{\mathbf{F}}^{-1}\bar{\mathbf{F}}^H\mathbf{b}\right)}{N-k+N\mathrm{tr}(\mathbf{\Phi}_{\mathbf{F}}^{-1})}. \tag{55}$$

Moreover, (54) and (55) are exact for $\bar{\mathbf{F}} = \mathbf{0}$, and are asymptotically exact for $\bar{\mathbf{F}} = \varrho\bar{\mathbf{F}}_0$ as $\varrho \to \infty$, where $\bar{\mathbf{F}}_0$ has full column rank. The right-hand sides of (54) and (55) lie in $[0, \|\mathbf{b}\|^2]$, as do their left-hand sides.

Proof: For brevity, let $\mathbf{\Phi} \triangleq \mathbf{\Phi_F}$. We first prove (54). Replacing $(\mathbf{F}^H\mathbf{F})^{-1}$ in the unnormalized ZF direction by $\mathbf{\Phi}^{-1} = (\mathbb{E}\{\mathbf{F}^H\mathbf{F}\})^{-1}$ yields $\bar{\mathbf{w}}_m(\mathbf{F}) \approx \mathbf{F}\mathbf{\Phi}^{-1}\mathbf{e}_m/\|\mathbf{F}\mathbf{\Phi}^{-1}\mathbf{e}_m\|$. Then, we approximate the mean of $|\mathbf{b}^H\mathbf{F}\mathbf{\Phi}^{-1}\mathbf{e}_m|^2/\|\mathbf{F}\mathbf{\Phi}^{-1}\mathbf{e}_m\|^2$ by the ratio of the means of its numerator and denominator, in the same spirit as [19, Lemma 1]. Since $\mathbb{E}\{\mathbf{W}\} = \mathbf{0}$ and $\mathbb{E}\{\mathbf{W}\mathbf{x}\mathbf{x}^H\mathbf{W}^H\} = \|\mathbf{x}\|^2\mathbf{I}_N$ for any deterministic $\mathbf{x}$, we have

$$\begin{aligned}\mathbb{E}\{|\mathbf{b}^H\mathbf{F}\mathbf{\Phi}^{-1}\mathbf{e}_m|^2\} &= |\mathbf{b}^H\bar{\mathbf{F}}\mathbf{\Phi}^{-1}\mathbf{e}_m|^2 + \|\mathbf{b}\|^2\|\mathbf{\Phi}^{-1}\mathbf{e}_m\|^2 \\ &= \left|[\mathbf{\Phi}^{-1}\bar{\mathbf{F}}^H\mathbf{b}]_m\right|^2 + \|\mathbf{b}\|^2[\mathbf{\Phi}^{-2}]_{m,m},\end{aligned} \tag{56}$$

$$\mathbb{E}\{\|\mathbf{F}\mathbf{\Phi}^{-1}\mathbf{e}_m\|^2\} = \mathbf{e}_m^T\mathbf{\Phi}^{-1}\mathbf{\Phi}\mathbf{\Phi}^{-1}\mathbf{e}_m = [\mathbf{\Phi}^{-1}]_{m,m}, \tag{57}$$

which lead to (54). Next, we prove (55). Replacing $(\mathbf{F}^H\mathbf{F})^{-1}$ in $\mathbf{P_F} \triangleq \mathbf{I}_N - \mathbf{P}_{\mathbf{F}}^{\perp} = \mathbf{F}(\mathbf{F}^H\mathbf{F})^{-1}\mathbf{F}^H$ by $\mathbf{\Phi}^{-1}$ and using $\mathbb{E}\{\mathbf{W}\mathbf{\Phi}^{-1}\mathbf{W}^H\} = \mathrm{tr}(\mathbf{\Phi}^{-1})\mathbf{I}_N$ yield

$$\mathbb{E}\{\mathbf{P_F}\} \approx \mathbf{\Pi} + \mathrm{tr}(\mathbf{\Phi}^{-1})\mathbf{I}_N, \qquad \mathbf{\Pi} \triangleq \bar{\mathbf{F}}\mathbf{\Phi}^{-1}\bar{\mathbf{F}}^H, \tag{58}$$

where $\mathbf{\Pi}$ and $\mathrm{tr}(\mathbf{\Phi}^{-1})\mathbf{I}_N$ represent the parts of the signal subspace induced by the LoS and NLoS components, respectively. However, (58) lets the NLoS component capture the fraction $\mathrm{tr}(\mathbf{\Phi}^{-1})$ of every direction, including those already captured by the LoS component, so that $\mathbf{b}^H(\mathbf{I}_N - \mathbf{\Pi} - \mathrm{tr}(\mathbf{\Phi}^{-1})\mathbf{I}_N)\mathbf{b}$ may even become negative when $\mathbf{b}$ is closely aligned with the column space of $\bar{\mathbf{F}}$. To avoid this double counting, we let the NLoS component capture a fraction $\vartheta \in [0,1]$ of the complement of $\mathbf{\Pi}$ only, i.e.,

$$\mathbb{E}\{\mathbf{P_F}\} \approx \mathbf{\Pi} + \vartheta(\mathbf{I}_N - \mathbf{\Pi}), \tag{59}$$

which satisfies $\mathbf{0} \preceq \mathbf{\Pi} + \vartheta(\mathbf{I}_N - \mathbf{\Pi}) \preceq \mathbf{I}_N$ since $\mathbf{0} \preceq \mathbf{\Pi} \prec \mathbf{I}_N$, as shown below. The fraction ϑ is determined by the identity $\mathrm{tr}(\mathbf{P_F}) = k$, which holds for every realization of $\mathbf{F}$. Since $\mathrm{tr}(\mathbf{\Pi}) = \mathrm{tr}\big(\mathbf{\Phi}^{-1}(\mathbf{\Phi} - N\mathbf{I}_k)\big) = k - N\mathrm{tr}(\mathbf{\Phi}^{-1})$, taking the trace of both sides of (59) yields $k - N\mathrm{tr}(\mathbf{\Phi}^{-1}) + \vartheta\big(N - k + N\mathrm{tr}(\mathbf{\Phi}^{-1})\big) = k$, i.e., $\vartheta = N\mathrm{tr}(\mathbf{\Phi}^{-1})/(N - k + N\mathrm{tr}(\mathbf{\Phi}^{-1}))$. Therefore, $\mathbb{E}\{\mathbf{P}_{\mathbf{F}}^{\perp}\} \approx (1-\vartheta)(\mathbf{I}_N - \mathbf{\Pi})$ with $1 - \vartheta = (N-k)/(N-k+N\mathrm{tr}(\mathbf{\Phi}^{-1}))$, which leads to (55).

Range: The nonzero eigenvalues of $\mathbf{\Pi}$ are $\mu_i^2/(\mu_i^2+N) < 1$, where $\mu_1, \ldots, \mu_k$ are the singular values of $\bar{\mathbf{F}}$, so that $\mathbf{0} \preceq \mathbf{\Pi} \prec \mathbf{I}_N$. Moreover, $\mathbf{\Phi} \succeq N\mathbf{I}_k$ implies $0 < \mathrm{tr}(\mathbf{\Phi}^{-1}) \leq k/N$, i.e., $1 - \vartheta \in [1 - k/N, 1)$. Hence, the right-hand side of (55) lies in $[0, \|\mathbf{b}\|^2]$. For (54), the Cauchy–Schwarz inequality and $N \geq 1$ imply that the numerator is at most $\|\mathbf{b}\|^2\mathbf{e}_m^T\mathbf{\Phi}^{-1}(\bar{\mathbf{F}}^H\bar{\mathbf{F}} + N\mathbf{I}_k)\mathbf{\Phi}^{-1}\mathbf{e}_m = \|\mathbf{b}\|^2[\mathbf{\Phi}^{-1}]_{m,m}$, so that the right-hand side of (54) lies in $[0, \|\mathbf{b}\|^2]$.

Exactness for $\bar{\mathbf{F}} = \mathbf{0}$: In this case, $\mathbf{\Phi} = N\mathbf{I}_k$ and $\mathbf{\Pi} = \mathbf{0}$, and the right-hand sides of (54) and (55) reduce to $\|\mathbf{b}\|^2/N$ and $\|\mathbf{b}\|^2(1 - k/N)$, respectively. Since $\mathbf{QW}$ has the same distribution as $\mathbf{W}$ for any unitary $\mathbf{Q}$, and $\bar{\mathbf{w}}_m(\mathbf{QW}) = \mathbf{Q}\bar{\mathbf{w}}_m(\mathbf{W})$, $\bar{\mathbf{w}}_m(\mathbf{W})$ is uniformly distributed on the unit sphere, and $\mathbb{E}\{\mathbf{P}_{\mathbf{W}}^{\perp}\}$ commutes with every unitary matrix and thus equals $(1-k/N)\mathbf{I}_N$ due to $\mathrm{tr}(\mathbf{P}_{\mathbf{W}}^{\perp}) = N - k$. Hence, the left-hand sides coincide with the right-hand sides.

Asymptotic exactness for $\bar{\mathbf{F}} = \varrho\bar{\mathbf{F}}_0$*:* As $\varrho \to \infty$, $\mathbf{F}/\varrho \to \bar{\mathbf{F}}_0$ almost surely. Since $\bar{\mathbf{w}}_m(\cdot)$ and $\mathbf{P}^{\perp}_{(\cdot)}$ are invariant to the scaling of their arguments and continuous at full-column-rank matrices, and $|\mathbf{b}^H\bar{\mathbf{w}}_m(\mathbf{F})|^2$ and $\mathbf{b}^H\mathbf{P}^{\perp}_{\mathbf{F}}\mathbf{b}$ are bounded by $\|\mathbf{b}\|^2$, the dominated convergence theorem yields $\mathbb{E}\{|\mathbf{b}^H\bar{\mathbf{w}}_m(\mathbf{F})|^2\} \to |\mathbf{b}^H\bar{\mathbf{w}}_m(\bar{\mathbf{F}}_0)|^2$ and $\mathbb{E}\{\mathbf{b}^H\mathbf{P}^{\perp}_{\mathbf{F}}\mathbf{b}\} \to \mathbf{b}^H\mathbf{P}^{\perp}_{\bar{\mathbf{F}}_0}\mathbf{b}$. On the other hand, $\varrho^2\mathbf{\Phi}^{-1} \to (\bar{\mathbf{F}}_0^H\bar{\mathbf{F}}_0)^{-1}$, $\mathbf{\Pi} \to \bar{\mathbf{F}}_0(\bar{\mathbf{F}}_0^H\bar{\mathbf{F}}_0)^{-1}\bar{\mathbf{F}}_0^H$, and $\mathrm{tr}(\mathbf{\Phi}^{-1}) \to 0$. Hence, the right-hand side of (55) converges to $\mathbf{b}^H\mathbf{P}^{\perp}_{\bar{\mathbf{F}}_0}\mathbf{b}$, and that of (54) converges to $|[(\bar{\mathbf{F}}_0^H\bar{\mathbf{F}}_0)^{-1}\bar{\mathbf{F}}_0^H\mathbf{b}]_m|^2/[(\bar{\mathbf{F}}_0^H\bar{\mathbf{F}}_0)^{-1}]_{m,m} = |\mathbf{b}^H\bar{\mathbf{w}}_m(\bar{\mathbf{F}}_0)|^2$, since the second term in the numerator of (54) decays as $\mathcal{O}(\varrho^{-4})$ whereas the denominator decays as $\mathcal{O}(\varrho^{-2})$. This completes the proof of Lemma 3. ∎

Proof of Proposition 1: From (15), $\bar{\mathbf{w}}_m = \bar{\mathbf{w}}_m(\mathbf{L}(\mathbf{t}) + \tilde{\mathbf{H}})$ and $\mathbf{P}^{\perp}(\mathbf{t}) = \mathbf{P}^{\perp}_{\mathbf{L}(\mathbf{t})+\tilde{\mathbf{H}}}$. Applying Lemma 3 with $\bar{\mathbf{F}} = \mathbf{L}(\mathbf{t})$, $\mathbf{W} = \tilde{\mathbf{H}}$, $k = M$, and $\mathbf{b} = \bar{\mathbf{g}}(\mathbf{t})$, for which $\mathbf{\Phi}_{\mathbf{F}} = \mathbf{\Phi}(\mathbf{t})$, $\bar{\mathbf{F}}^H\mathbf{b} = \mathbf{v}(\mathbf{t})$, and $\|\mathbf{b}\|^2 = N$, yields $\mathbb{E}\{x_m\} \approx c_m(\mathbf{t})$ and $\mathbb{E}\{y\} \approx d(\mathbf{t})$. Substituting them into (21) yields (24), which completes the proof.

APPENDIX B
DERIVATIVES OF ω_m, c_m, AND d

Let $o_m \triangleq [\mathbf{\Omega}^2]_{m,m}$, $\tau \triangleq \mathrm{tr}(\mathbf{\Omega})$, $\tilde{\mathbf{v}} \triangleq \mathbf{\Omega}\mathbf{v}$, and $\varpi \triangleq \mathbf{v}^H\mathbf{\Omega}\mathbf{v}$. Since $\mathbf{\Omega}$ is Hermitian, (46) can be rewritten as $\dot{\mathbf{\Omega}} = -(\mathbf{\Omega}\dot{\mathbf{l}}_n)(\mathbf{\Omega}\mathbf{l}_n)^H - (\mathbf{\Omega}\mathbf{l}_n)(\mathbf{\Omega}\dot{\mathbf{l}}_n)^H$. Hence, the derivatives of ω_m, o_m, and τ w.r.t. $u \in \{x_n, y_n\}$ are given by

$$\dot{\omega}_m = [\dot{\mathbf{\Omega}}]_{m,m} = -2\mathrm{Re}\big\{[\mathbf{\Omega}\dot{\mathbf{l}}_n]_m[\mathbf{\Omega}\mathbf{l}_n]^*_m\big\}, \tag{60}$$

$$\dot{o}_m = 2\mathrm{Re}\big\{[\mathbf{\Omega}\dot{\mathbf{\Omega}}]_{m,m}\big\} = -2\mathrm{Re}\big\{[\mathbf{\Omega}^2\dot{\mathbf{l}}_n]_m[\mathbf{\Omega}\mathbf{l}_n]^*_m + [\mathbf{\Omega}^2\mathbf{l}_n]_m[\mathbf{\Omega}\dot{\mathbf{l}}_n]^*_m\big\}, \tag{61}$$

$$\dot{\tau} = \mathrm{tr}(\dot{\mathbf{\Omega}}) = -2\mathrm{Re}\big\{\mathbf{l}_n^H\mathbf{\Omega}^2\dot{\mathbf{l}}_n\big\}, \tag{62}$$

where (61) follows from $[\dot{\mathbf{\Omega}}\mathbf{\Omega}]_{m,m} = [\mathbf{\Omega}\dot{\mathbf{\Omega}}]^*_{m,m}$. For c_m in (22), the derivative of $\tilde{\mathbf{v}} = \mathbf{\Omega}\mathbf{v}$ is

$$\dot{\tilde{\mathbf{v}}} = \dot{\mathbf{\Omega}}\mathbf{v} + \mathbf{\Omega}\dot{\mathbf{v}} = -\mathbf{\Omega}\dot{\mathbf{l}}_n\big(\mathbf{l}_n^H\tilde{\mathbf{v}}\big) - \mathbf{\Omega}\mathbf{l}_n\big(\dot{\mathbf{l}}_n^H\tilde{\mathbf{v}}\big) + \mathbf{\Omega}\dot{\mathbf{v}}, \tag{63}$$

and thus

$$\dot{c}_m = \frac{2\mathrm{Re}\big\{[\tilde{\mathbf{v}}]^*_m[\dot{\tilde{\mathbf{v}}}]_m\big\} + N\dot{o}_m - c_m\dot{\omega}_m}{\omega_m}. \tag{64}$$

For $d = (1-\vartheta)(N-\varpi)$ in (23), where $1-\vartheta = (N-M)/(N-M+N\tau)$ as in Appendix A, since $\mathbf{v}^H\dot{\mathbf{\Omega}}\mathbf{v} = -2\mathrm{Re}\{(\tilde{\mathbf{v}}^H\dot{\mathbf{l}}_n)(\mathbf{l}_n^H\tilde{\mathbf{v}})\}$, we have

$$\dot{\varpi} = 2\mathrm{Re}\{\tilde{\mathbf{v}}^H\dot{\mathbf{v}}\} - 2\mathrm{Re}\big\{(\tilde{\mathbf{v}}^H\dot{\mathbf{l}}_n)(\mathbf{l}_n^H\tilde{\mathbf{v}})\big\}, \tag{65}$$

$$\dot{d} = -(1-\vartheta)\left(\dot{\varpi} + \frac{N(N-\varpi)\,\dot{\tau}}{N-M+N\tau}\right). \tag{66}$$

Substituting $\dot{\eta}_m = -\eta_m\dot{\omega}_m/\omega_m$, (64), and (66) into (47) yields $\nabla_{\mathbf{t}_n}F$. All the quantities in (60)–(66) are obtained from matrix-vector products with $\mathbf{\Omega}$, which require $\mathcal{O}(M^2)$ operations.

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